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Analysis of a Stochastic Within-Host Model of Dengue Infection with Immune Response and Ornstein–Uhlenbeck Process

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Abstract
In this paper, assuming the certain variable satisfies the Ornstein–Uhlenbeck process, we formulate a stochastic within-host dengue model with immune response to obtain further understanding of the transmission dynamics of dengue fever. Then we analyze the dynamical properties of the stochastic system in detail, including the existence and uniqueness of the global solution, the existence of a stationary distribution, and the extinction of infected monocytes and free viruses. In particular, it is worth revealing that we get the specific form of covariance matrix in its probability density around the quasi-endemic equilibrium of the stochastic system. Finally, numerical illustrative examples are depicted to confirm our theoretical findings.

Keywords Dengue model · Immune response · Ornstein–Uhlenbeck process · Stationary distribution · Probability density · Extinction

Mathematics Subject Classification 34E10 · 60H10 · 92B05 · 92D30

1 Introduction

Dengue fever (DF) is a mosquito-borne viral infection caused by one of the four dengue virus stereotypes (DENV-1 to DENV-4) (Kiouach et al. 2023). It is primarily transmitted to humans by mosquitoes such as Aedes aegypti and Aedes albopictus (Zha and

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Jiang 2023; Gubler 1998; Hales et al. 2002). Nowadays, more than fifty-five percent of the world's population live in areas at risk of dengue fever transmission (Sharma et al. 2021; Abidemi et al. 2022), placing almost 3.9 billion people at risk of infection (Abidemi et al. 2022; Tamura et al. 2021). According to the World Health Organization (WHO) (WHO 2020), a large number of dengue infections produce mild illness, whereas many others lead to an acute flu-like illness named severe dengue (Kiouach et al. 2023). Until now, although enormous efforts are in place to develop an actual vaccine, there is no effective treatment for dengue fever or commercial vaccine due to antibody-dependent enhancement between different serotypes (Uno and Ross 2021; Wang et al. 2022). In this context, many mathematicians have proposed various models to study different aspects of dengue for better describing the dengue fever prevalence and perceiving its influences on a host population (Zha and Jiang 2023; Sharma et al. 2021; Abidemi et al. 2022; Wang et al. 2022; Kim et al. 2020; Jan et al. 2020; Ali et al. 2022; Abidemi et al. 2021; Xue et al. 2021; Abidemi et al. 2020a; Brito da Cruz and Rodrigues 2021; Abidemi et al. 2020b; Nuraini et al. 2009). These models include the impacts of human movements on the spread of disease in a patchy environment (Kim et al. 2020; Abidemi et al. 2021), coexistence of two or more dengue virus serotypes (Xue et al. 2021; Abidemi et al. 2020a; Brito da Cruz and Rodrigues 2021), effects of seasonal variation (Kim et al. 2020; Abidemi et al. 2020b) and within-host dengue virus transmission dynamics (Sharma et al. 2021). In particular, Nuraini et al. (2009) developed a within-host dengue viral infection model with immune response which is governed by the following ordinary differential equations system:

$$\begin{cases} \dot{S} = \mu - \alpha S - bSV, \\ \dot{I} = bSV - \beta I - \nu IZ, \\ \dot{V} = mI - \gamma V - bSV, \\ \dot{Z} = \eta + kI + \theta IZ - \delta Z, \end{cases} \tag{1.1}$$

where S denotes the susceptible monocyte population, I denotes the infected monocyte population, V denotes the viral load and Z denotes the immune cells. All parameters are assumed to be positive constants, and their descriptions are presented in Table 1.

By using the next-generation matrix approach (Van den Driessche and Watmough 2002), the authors obtained the basic reproduction number of model (1.1) which takes the following form

$$\mathcal{R}_0 = \sqrt{\frac{mb\mu\delta}{(\alpha\gamma + b\mu)(\beta\delta + \nu\eta)}},$$

and it is used to determine whether the virus disappears or not. In addition, the dynamics of model (1.1) are presented as follows:

- If $\mathcal{R}_0 < 1$, the disease-free equilibrium $E_0 = (S^0, 0, 0, Z^0)$ always exists and it is locally asymptotically stable in the invariant set Γ, where

$$\Gamma = \Big\{ (S, I, V, Z) | 0 < S \leq S^0, 0 < S + I \leq \tfrac{\mu}{\min\{\alpha,\beta\}}, \\ 0 \leq V \leq \tfrac{m\mu}{\gamma \min\{\alpha,\beta\}}, Z^0 \leq Z, 0 < I + V + \tfrac{\nu}{\theta} Z$$

$$\leq \frac{\nu\eta \min\{\alpha,\beta\}+\mu(m\theta+k\nu)}{\theta \min\{\alpha,\beta\} \min\{\beta,\gamma,\delta\}}\Bigg\},\ S^0 = \frac{\mu}{\alpha},\ Z^0 = \frac{\eta}{\delta}.$$

- If $\mathcal{R}_0 > 1$, E_0 is unstable and there is also a unique endemic equilibrium $E^+ = (S^+, I^+, V^+, Z^+)$, where

$$S^+ = \frac{\mu}{\alpha + bV^+},\ I^+ = \frac{\gamma V^+(\alpha + bV^+) + b\mu V^+}{m(\alpha + bV^+)},$$
$$Z^+ = \frac{m\alpha\eta + V^+(mb\eta + k\alpha\gamma + kb\mu) + kb\gamma(V^+)^2}{m\alpha\delta + V^+(mb\delta - \theta\alpha\gamma - b\theta\mu) - b\theta\gamma(V^+)^2},$$

  and V^+ is the positive solution of the following algebraic equation:

$$(m-\beta)b\mu - \beta\gamma(\alpha + bV) - \frac{[\gamma\nu(\alpha + bV) + b\mu\nu][m\alpha\eta + V(mb\eta + k\alpha\gamma + kb\mu) + kb\gamma V^2]}{m\alpha\delta + V(mb\delta - \theta\alpha\gamma - b\theta\mu) - b\theta\gamma V^2} = 0.$$

Although model (1.1) is well-established and easily understood, while ignoring stochastic variation of the system, it is very common that the transmission of dengue fever from mosquitoes to humans involves many stochastic factors, such as climate, exposure to mosquitoes, migration of humans and mosquitoes, behavior changes of humans, etc. (Sun et al. 2018). In other terms, and to be more explicit, it is urgent to formulate stochastic models which are differentiated from their deterministic counterparts since when the model has studied for the same values of parameters and initial conditions, the stochastic model may produce different outputs compared to the same deterministic model (Keeling and Rohani 2008). In this regard, many scholars have proposed and developed some stochastic models that are presenting the dengue infection dynamics from different perspectives (Kiouach et al. 2023; Sun et al. 2018;

Table 1 Summary of parameter meaning of system (1.1)

Parameters	Descriptions
μ	Production rate of susceptible monocytes
b	Contact rate between susceptible monocytes and viral load
ν	Clearance rate of infected monocytes
θ	Contact rate between infected monocytes and immune cells
α	Natural death rate of susceptible monocytes
β	Natural death rate of infected monocytes
m	Multiplication of virus
γ	Natural death rate of viral load
η	Production rate of immune cells
k	Proportionality factor of infected monocytes
δ	Natural death rate of immune cells

Chang and Zhang 2022; Otero and Solari 2010; Liu and Din 2021; Liu et al. 2019, 2018) because stochastic models can better account for demographic or environmental randomness.

When building up stochastic models, parameter perturbation technology is considered to be the most common way to introduce stochastic noise in biologically realistic dynamic models. So far, there are two main parameter perturbation ways to introduce stochastic perturbations in deterministic models. One way is to incorporate linear white noise in the deterministic system, which has been widely employed in the existing references (Liu and Din 2021; Liu et al. 2018; Qi et al. 2021; Xu et al. 2021; Zhao et al. 2020; Tan et al. 2022; Hussain et al. 2022). Inspired by this idea, we suppose that $b(t)$ is affected by the following linear white noise

$$b(t) = \bar{b} + \frac{\sigma \mathrm{d}B(t)}{\mathrm{d}t}, \tag{1.2}$$

where $B(t)$ is a standard Brownian motion defined on a complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ with a filtration $\{\mathcal{F}_t\}_{t\geq 0}$ satisfying the usual conditions (Mao 1997). $\sigma^2 > 0$ is the noise intensity. However, this type of approach has two drawbacks. On the one hand, (1.2) cannot ensure the nonnegativity of the contact rate $b(t)$. On the other hand, if we use the notation $\langle b(t)\rangle$ to represent the time average of $b(t)$ on the interval $[0, t]$, then

$$\langle b(t)\rangle = \frac{1}{t}\int_0^t b(s)\mathrm{d}s = \bar{b} + \frac{\sigma B(t)}{t} \sim \mathbb{N}\left(\bar{b}, \frac{\sigma^2}{t}\right).$$

Obviously, the variance σ^2/t of $\langle b(t)\rangle$ on the interval $[0, t]$ will tend to infinity as $t \to 0^+$. This is inconsistent with facts because the fluctuation of $b(t)$ will become very large in sufficiently small interval.

The other way is to suppose that each random variable is determined by an Ornstein–Uhlenbeck process. Recently, there have been a lot of excellent works to illustrate the advantages of Ornstein–Uhlenbeck process over linear white noise (Liu 2023a; Zhou and Jiang 2023; Shi and Jiang 2022; Liu 2023b; Allen 2016; Zhou et al. 2022). If we disturb the contact rate b directly with the Ornstein–Uhlenbeck process, then it may be negative due to the characteristics of this type of process. That is to say, its nonnegativity cannot be ensured either. Motivated by the works of Allen (2016), we suppose that the variable $\ln b$ is affected by an Ornstein–Uhlenbeck process, and it is described by the following stochastic differential equation:

$$\mathrm{d}(\ln b) = \rho(\ln \bar{b} - \ln b)\mathrm{d}t + \sigma \mathrm{d}B(t), \tag{1.3}$$

where $\ln \bar{b}$ is the long-term mean of the process, ρ is the speed of reversion, and σ and $B(t)$ are the same as in (1.2).

It is easy to see that system (1.3) has a unique exact solution

$$\ln b(t) = \ln \bar{b} + (\ln b(0) - \ln \bar{b})e^{-\rho t} + \sigma \int_0^t e^{-\rho(t-s)}\mathrm{d}B(s),$$

which shows that the variable $\ln b(t) \sim \mathbb{N}(\ln \bar{b} + (\ln b(0) - \ln \bar{b})e^{-\rho t}, \sigma^2(1 - e^{-2\rho t})/(2\rho))$. It is noted that

$$\lim_{t\to 0^+} \mathbb{E}(\ln b(t)) = \ln b(0), \quad \lim_{t\to 0^+} \mathbf{Var}(\ln b(t)) = 0, \quad \lim_{t\to\infty} \mathbb{E}(\ln b(t)) = \ln \bar{b},$$
$$\lim_{t\to\infty} \mathbf{Var}(\ln b(t)) = \frac{\sigma^2}{2\rho}.$$

In addition, for any interval $[0, t]$, we have

$$\mathbf{Var}(\langle \ln b(t) \rangle) = \frac{\sigma^2}{3} t + O(t^2).$$

This implies that the fluctuation of $\ln b(t)$ will be small enough in a small interval, which is consistent with facts. Consequently, this approach is reasonable to simulate the random impacts of key parameters both from biological significance and mathematical perspective.

Actually, the main aim of this paper is to analyze the transmission dynamics of dengue infection, that is, the dynamical properties of $S(t)$, $I(t)$, $V(t)$ and $Z(t)$. For this purpose, we let $x = \ln b$, then the stochastic form corresponding to system (1.1) can be given as follows:

$$\begin{cases} \mathrm{d}S = (\mu - \alpha S - e^x SV)\mathrm{d}t, \\ \mathrm{d}I = (e^x SV - \beta I - \nu IZ)\mathrm{d}t, \\ \mathrm{d}V = (mI - \gamma V - e^x SV)\mathrm{d}t, \\ \mathrm{d}Z = (\eta + kI + \theta IZ - \delta Z)\mathrm{d}t, \\ \mathrm{d}x = \rho(\bar{x} - x)\mathrm{d}t + \sigma \mathrm{d}B(t). \end{cases} \tag{1.4}$$

The main purpose of this paper is to study the existence of a stationary distribution theoretically, as well as the specific form of covariance matrix in its probability density near the quasi-endemic equilibrium of system (1.4). As we all know, few works have studied the stochastic models with a biologically reasonable stochastic process, i.e., the Black–Karasinski process. There are two obstacles listed as follows.

- On the one hand, in the investigation of disease transmission, an interesting and important topic is whether dengue fever can coexist with human beings, that is, the existence of a stationary distribution of system (1.4). The approach used here is the Lyapunov function method, which is widely employed by many scholars (see, for example, Liu 2023a; Zhou and Jiang 2023; Shi and Jiang 2022; Liu 2023b; Zhou et al. 2022, 2023; Han et al. 2020) to get the global dynamical behaviors of some viral infection models. Nevertheless, there is usually no general method to construct such appropriate Lyapunov functions. Thus, we need to construct some new forms of nonnegative Lyapunov functions which are independent of the positive equilibria of their corresponding deterministic forms to overcome the obstacle.
- On the other hand, on the basis of the existence of a stationary distribution, we further hope to obtain the specific form of covariance matrix of the density

function corresponding to this stationary distribution. However, our model is five-dimensional, which is a high-dimensional system. Hence, it is difficult to get the form of covariance matrix. To this end, new mathematical techniques and more delicate analysis need to be developed to make it.

In general, it is a difficult problem for us to carry out a complete dynamic analysis of the system (1.4) urgently. In comparison with the works done on dengue infection models in the existing literature, the contributions and innovations of this paper can be stated as follows:

- In this work, we adopt the Ornstein–Uhlenbeck process in logarithmic form to establish a stochastic model which is used to study the transmission dynamics of dengue virus infection. This process is more reasonable to simulate the random effects than the linear white noise perturbation. Our model may reflect more reality.
- We adopt new mathematical techniques and carry out more delicate analysis to get two stochastic parameters $\mathcal{R}_0^S$ and $\mathcal{R}_0^E$ which are used to determine the extinction and persistence of infected monocytes and free viruses in the stochastic system (1.4). More precisely, if $\mathcal{R}_0^S > 1$, then system (1.4) has at least one stationary distribution $\pi(\cdot)$ which shows that the dengue viruses will be persistent in the humans while if $\mathcal{R}_0^E < 1$ and $\mathcal{R}_0 > \bar{b}S^0/(\gamma + \bar{b}S^0)$, then the infected monocytes and free viruses will go to extinction exponentially with probability one. In addition, if $\sigma = 0$, then $\mathcal{R}_0^S = \mathcal{R}_0^E = \mathcal{R}_0$, which implies that our results are a good generalization of the deterministic system (1.1).
- By solving the five-dimensional algebraic equations related to the stochastic system (1.4), we obtain the specific form of covariance matrix in its probability density, which is a function that describes the probability of the output value of the random variable near the quasi-endemic equilibrium of system (1.4). Mathematically, the existence of probability density of system (1.4) is more in-depth and specific than that of the stationary distribution.
- It is worth pointing out that the mathematical techniques and analysis methods used in this paper can be applied to other stochastic complicated epidemic models with the Ornstein–Uhlenbeck process in logarithmic form.

Of course, there are also some distinct limitations in our work. Firstly, in this paper, we only obtain the existence of a stationary distribution without obtaining its uniqueness and ergodicity. Secondly, we only derive sufficient criteria for exponential extinction of infected monocytes and free viruses, but how to make the criteria for determining the persistence and extinction of stochastic viral infection systems consistent or close enough is unsolved. Thirdly, as we know, the actual biological dynamics themselves may be spatially heterogeneous. For example, in the process of vegetation evolution in arid systems, vegetation in different areas may exhibit different patterns due to differences in rainfall and runoff influences (Chen et al. 2021; Kefi et al. 2008). Thus, we can consider the influences of spatial diffusion factors. Finally, we would like to reveal that due to the unknown coefficients in the model, for practical applications in epidemiological dynamics, we need to adopt statistical methods to estimate the main parameters involved in stochastic models. We will consider these issues in our future work. There are still many limitations in the paper, and we won't list them all here.

We arrange the rest part of this article in the way as follows. In Sect. 2, the existence and uniqueness of the global solution of the stochastic system (1.4) is proved which is important to analyze the dynamics of a within-host model. In Sect. 3, we employ new techniques to develop several Lyapunov functions to get sufficient criteria for the existence of a stationary distribution of system (1.4). In Sect. 4, under some appropriate conditions, we get the exact form of covariance matrix in the probability density around the quasi-endemic equilibrium E^* of system (1.4). In Sect. 5, we get sufficient criteria for extinction of the infected monocytes and free viruses. In Sect. 6, through the numerical simulations of several examples, we verify the effectiveness of our theoretical findings. Finally, we present a concise conclusion to summarize our theoretical results.

Throughout this paper, for the sake of convenience, we give the following notations:

$$\begin{aligned}
&\mathbb{R}_+^d = \{x = (x_1, \ldots, x_d) \in \mathbb{R}^d : x_i > 0, 1 \le i \le d\} \\
&\qquad \text{and } \overline{\mathbb{R}}_+^d = \{x = (x_1, \ldots, x_d) \in \mathbb{R}^d : x_i \ge 0, 1 \le i \le d\}, \\
&x \wedge y = \min\{x, y\} \text{ for any } x, y \in \mathbb{R}.
\end{aligned}$$

Let $C^{2,1}(\mathbb{R}^d \times \mathbb{R}_+; \overline{\mathbb{R}}_+)$ be the family of all nonnegative functions $V(X, t)$ defined on $\mathbb{R}^d \times \mathbb{R}_+$ such that they are continuously twice differentiable in X and once in t. Let $\mathbf{1}_{\mathbb{A}}$ be the indicator function of the set $\mathbb{A}$. If G is a vector, the notation G^T is used to represent its transpose. If G is an invertible matrix, its inverse matrix is expressed by G^{-1}. If G is a square matrix, the notation $|G|$ is used to represent its determinant. The notation $\mathbb{G}^{(k)}$ is used to denote the kth leading principal minor of the matrix $\mathbb{G}$.

Next, we will present some fundamental theories on stochastic differential equations. (For more details, we refer the readers to Mao 1997.)

Let $X(t)$ be a d-dimensional stochastic process on $t \ge 0$ which is given by the following stochastic differential equation:

$$\mathrm{d}X(t) = f(X(t), t)\mathrm{d}t + g(X(t), t)\mathrm{d}B(t),$$

where $f \in L^1(\mathbb{R}_+; \mathbb{R}^d)$, $g \in L^2(\mathbb{R}_+; \mathbb{R}^{d\times m})$ and $B(t) = \{(B_t^1, B_t^2, \ldots, B_t^d)^T\}_{t\ge 0}$ is a d-dimensional Brownian motion defined on the complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\ge 0}, \mathbb{P})$. We define the differential operator L as follows:

$$L = \frac{\partial}{\partial t} + \sum_{i=1}^{d} f_i(X(t), t)\frac{\partial}{\partial X_i} + \frac{1}{2}\sum_{i,j=1}^{d} (g^T(X(t), t)g(X(t), t))_{ij}\frac{\partial^2}{\partial X_i \partial X_j}.$$

If L acts on a function $V \in C^{2,1}(\mathbb{R}^d \times \mathbb{R}_+; \mathbb{R})$, then

$$\begin{aligned}
LV(X(t), t) = {} & V_t(X(t), t) + V_X(X(t), t)f(X(t), t) \\
& + \frac{1}{2}\mathrm{trace}[g^T(X(t), t)V_{XX}(X(t), t)g(X(t), t)],
\end{aligned}$$

where $V_t = \partial V/\partial t$, $V_X = (\partial V/\partial X_1, \ldots, \partial V/\partial X_d)$ and $V_{XX} = (\partial^2 V/(\partial X_i \partial X_j))_{d\times d}$.

2 Existence and Uniqueness of the Global Solution

One of the main aims in studying the transmission dynamics of a viral infection system is that whether the solution of the system is global or not. To this end, we first present a theorem which is related to the existence and uniqueness of the global solution of system (1.4).

Theorem 2.1 *For any initial value* $(S(0), I(0), V(0), Z(0), x(0))^T \in \mathbb{R}_+^4 \times \mathbb{R}$*, system* (1.4) *has a unique solution* $(S(t), I(t), V(t), Z(t), x(t))^T$ *on* $t \geq 0$ *and the solution will remain in* $\mathbb{R}_+^4 \times \mathbb{R}$ *with probability one.*

Proof Since all the coefficients of system (1.4) satisfy the local Lipschitz conditions, then for any initial value $(S(0), I(0), V(0), Z(0), x(0))^T \in \mathbb{R}_+^4 \times \mathbb{R}$ there is a unique maximum local solution $(S(t), I(t), V(t), Z(t), x(t))^T$ on the interval $[0, \tau_e)$, where τ_e is an explosion time (Mao 1997). Now we prove this solution is global, that is to say, we need to prove $\tau_e = \infty$ a.s. For this reason, let $n_0 \geq 1$ be large enough such that $S(0)$, $I(0)$, $V(0)$, $Z(0)$ and $e^{x(0)}$ all lie within the interval $[1/n_0, n_0]$. For each integer $n \geq n_0$, we define a stopping time by Mao (1997)

$$\begin{aligned}\tau_n = \inf \Big\{ & t \in [0, \tau_e) : \min\{S(t), I(t), V(t), Z(t), e^{x(t)}\} \\ & \leq \frac{1}{n} \text{ or } \max\{S(t), I(t), V(t), Z(t), e^{x(t)}\} \geq n \Big\},\end{aligned}$$

where throughout this paper we set $\inf \emptyset = \infty$ (here $\emptyset$ is the empty set). One can easily see that the stopping time τ_n is increasing as $n \to \infty$. Denote by $\tau_\infty = \lim_{n\to\infty} \tau_n$, then $\tau_\infty \leq \tau_e$ a.s. If $\tau_\infty = \infty$ a.s. is true, we get $\tau_e = \infty$ a.s. and $(S(t), I(t), V(t), Z(t), x(t))^T \in \mathbb{R}_+^4 \times \mathbb{R}$ a.s. for any $t \geq 0$. In other words, to accomplish the proof we only need to verify $\tau_\infty = \infty$ a.s. If this assertion is false, then there are two constants $T > 0$ and $\epsilon \in (0, 1)$ such that

$$\mathbb{P}\{\tau_\infty \leq T\} > \epsilon.$$

As a result, there is an integer $n_1 \geq n_0$ such that

$$\mathbb{P}\{\tau_n \leq T\} \geq \epsilon \text{ for any } n \geq n_1.$$

Next, let us define a C^2-function $\mathcal{U} : \mathbb{R}_+^4 \times \mathbb{R} \to \overline{\mathbb{R}}_+$ by

$$\begin{aligned}\mathcal{U}(S, I, V, Z, x) = & (S - 1 - \ln S) + (I - 1 - \ln I) + (V - 1 - \ln V) \\ & + \frac{\nu}{\theta}(Z - 1 - \ln Z) + (e^x - 1 - x).\end{aligned}$$

It is noted that the nonnegativity of this function can be ensured due to the inequality $u - 1 - \ln u \geq 0$ for any $u > 0$. Applying Itô's formula (Mao 1997) to $\mathcal{U}$, one has

$$\mathrm{d}\mathcal{U}(S, I, V, Z, x) = L\mathcal{U}(S, I, V, Z, x)\mathrm{d}t + \sigma(e^x - 1)\mathrm{d}B(t),$$

where $L\mathcal{U} : \mathbb{R}_+^4 \times \mathbb{R} \to \mathbb{R}$ is defined by

$$\begin{aligned} L\mathcal{U} &= \mu - \alpha S - e^x SV - \frac{\mu}{S} + \alpha + e^x V + e^x SV - \beta I - \nu IZ \\ &\quad - \frac{e^x SV}{I} + \beta + \nu Z + mI - \gamma V - e^x SV - \frac{mI}{V} \\ &\quad + \gamma + e^x S + \frac{\nu\eta}{\theta} + \frac{k\nu}{\theta} I + \nu IZ - \frac{\nu\delta}{\theta} Z - \frac{\nu\eta}{\theta Z} - \frac{k\nu I}{\theta Z} - \nu I \\ &\quad + \frac{\nu\delta}{\theta} + \rho\bar{x}e^x + \rho x - \rho x e^x - \rho\bar{x} + \frac{\sigma^2}{2} e^x \\ &\le \mu + \frac{\nu\eta}{\theta} + \alpha + \beta + \gamma + \frac{\nu\delta}{\theta} - \rho\bar{x} + \left(m + \frac{k\nu}{\theta}\right) I + \nu Z \\ &\quad + e^x S + e^x V + \rho\bar{x}e^x + \rho x - \rho x e^x + \frac{\sigma^2}{2} e^x. \end{aligned} \tag{2.1}$$

In addition, in view of system (1.4), one gets

$$\begin{aligned} \mathrm{d}S &= (\mu - \alpha S - e^x SV)\mathrm{d}t \\ &\le (\mu - \alpha S)\mathrm{d}t, \end{aligned}$$

and

$$\begin{aligned} \mathrm{d}(S + I) &= (\mu - \alpha S - \beta I - \nu IZ)\mathrm{d}t \\ &\le [\mu - (\alpha \wedge \beta)(S + I)]\mathrm{d}t, \end{aligned}$$

which shows that

$$S(t) \le \begin{cases} S(0), & \text{if } S(0) > \dfrac{\mu}{\alpha}, \\ \dfrac{\mu}{\alpha}, & \text{if } S(0) \le \frac{\mu}{\alpha} \end{cases} \le K_1, \tag{2.2}$$

and

$$S(t) + I(t) \le \begin{cases} S(0) + I(0), & \text{if } S(0) + I(0) > \dfrac{\mu}{\alpha \wedge \beta}, \\ \dfrac{\mu}{\alpha \wedge \beta}, & \text{if } S(0) + I(0) \le \frac{\mu}{\alpha \wedge \beta} \end{cases} \le K_2, \tag{2.3}$$

where

$$K_1 := \max\left\{S(0), \frac{\mu}{\alpha}\right\}, \ K_2 := \max\left\{S(0) + I(0), \frac{\mu}{\alpha \wedge \beta}\right\}.$$

On the other hand, one obtains

$$\begin{aligned} \mathrm{d}V &= (mI - \gamma V - e^x SV)\mathrm{d}t \\ &\le (mI - \gamma V)\mathrm{d}t \\ &\le (mK_2 - \gamma V)\mathrm{d}t, \end{aligned}$$

and

$$\begin{aligned} d\left(I + V + \frac{\nu}{\theta}Z\right) &= \frac{\nu\eta}{\theta} + mI + \frac{k\nu}{\theta}I - \beta I - \gamma V - \frac{\nu\delta}{\theta}Z \\ &\le \frac{\nu\eta}{\theta} + \left(m + \frac{k\nu}{\theta}\right)K_2 - (\beta \wedge \gamma \wedge \delta)\left(I + V + \frac{\nu}{\theta}Z\right), \end{aligned}$$

which indicates that

$$V(t) \le \begin{cases} V(0), & \text{if } V(0) > \dfrac{mK_2}{\gamma}, \\ \dfrac{mK_2}{\gamma}, & \text{if } V(0) \le \frac{mK_2}{\gamma} \end{cases} \le K_3, \tag{2.4}$$

and

$$\begin{aligned} &I(t) + V(t) + \frac{\nu}{\theta}Z(t) \\ &\le \begin{cases} I(0) + V(0) + \dfrac{\nu}{\theta}Z(0), & \text{if } I(0) + V(0) + \frac{\nu}{\theta}Z(0) > \dfrac{\nu\eta + K_2(m\theta + k\nu)}{\theta(\beta \wedge \gamma \wedge \delta)}, \\ \dfrac{\nu\eta + K_2(m\theta + k\nu)}{\theta(\beta \wedge \gamma \wedge \delta)}, & \text{if } I(0) + V(0) + \frac{\nu}{\theta}Z(0) \le \frac{\nu\eta + K_2(m\theta + k\nu)}{\theta(\beta \wedge \gamma \wedge \delta)} \end{cases} \le K_4, \end{aligned} \tag{2.5}$$

where

$$\begin{aligned} K_3 &:= \max\left\{V(0), \frac{mK_2}{\gamma}\right\}, \\ K_4 &:= \max\left\{I(0) + V(0) + \frac{\nu}{\theta}Z(0), \frac{\nu\eta + K_2(m\theta + k\nu)}{\theta(\beta \wedge \gamma \wedge \delta)}\right\}. \end{aligned}$$

When (2.2), (2.3), (2.4) and (2.5) are used in (2.1), one has

$$\begin{aligned} L\mathcal{U} &\le \mu + \frac{\nu\eta}{\theta} + \alpha + \beta + \gamma + \frac{\nu\delta}{\theta} - \rho\bar{x} + \left(m + \frac{k\nu}{\theta}\right)K_2 \\ &\quad + K_4\theta + K_1e^x + K_3e^x + \rho\bar{x}e^x + \rho x - \rho xe^x + \frac{\sigma^2}{2}e^x \\ &:= \mu + \frac{\nu\eta}{\theta} + \alpha + \beta + \gamma + \frac{\nu\delta}{\theta} - \rho\bar{x} + K_2\left(m + \frac{k\nu}{\theta}\right) + K_4\theta + f(x), \end{aligned}$$

where

$$f(x) = -\rho x e^x + \rho x + \left(\rho\bar{x} + K_1 + K_3 + \frac{\sigma^2}{2}\right)e^x.$$

Note that $f(-\infty) = f(\infty)$. As a result, on the real number domain, the function $f(x)$ is upper bounded. Then

$$\begin{aligned} L\mathcal{U} &\le \mu + \frac{\nu\eta}{\theta} + \alpha + \beta + \gamma + \frac{\nu\delta}{\theta} - \rho\bar{x} + K_2\left(m + \frac{k\nu}{\theta}\right) + K_4\theta + \sup_{x\in\mathbb{R}}\{f(x)\} \\ &:= K_5. \end{aligned}$$

Here K_5 is a constant which is independent of the variables S, I, V, Z and x. The rest of the proof is similar to that of Zhou et al. (2022), and we omit it here. This completes the proof. □

Remark 2.1 By Theorem 2.1, it is clear that for any initial value $(S(0), I(0), V(0), Z(0), x(0))^T \in \mathbb{R}_+^4 \times \mathbb{R}$, system (1.4) has a unique global solution $(S(t), I(t), V(t), Z(t), x(t))^T$ on $t \ge 0$ with probability one.

In view of system (1.4), one gets

$$\begin{aligned} \mathrm{d}S &= (\mu - \alpha S - e^x SV)\mathrm{d}t \\ &\le (\mu - \alpha S)\mathrm{d}t, \\ \mathrm{d}(S+I) &= (\mu - \alpha S - \beta I - \nu I Z)\mathrm{d}t \\ &\le [\mu - (\alpha\wedge\beta)(S+I)]\mathrm{d}t, \\ \mathrm{d}V &= (mI - \gamma V - e^x SV)\mathrm{d}t \\ &\le (mI - \gamma V)\mathrm{d}t, \\ \mathrm{d}Z &= (\eta + kI + \theta I Z - \delta Z)\mathrm{d}t \\ &\ge (\eta - \delta Z)\mathrm{d}t, \end{aligned}$$

and

$$\begin{aligned} d\left(I + V + \frac{\nu}{\theta}Z\right) &= \frac{\nu\eta}{\theta} + mI + \frac{k\nu}{\theta}I - \beta I - \gamma V - \frac{\nu\delta}{\theta}Z \\ &\le \frac{\nu\eta}{\theta} + \left(m + \frac{k\nu}{\theta}\right)I - (\beta\wedge\gamma\wedge\delta)\left(I + V + \frac{\nu}{\theta}Z\right). \end{aligned}$$

Consequently, we have

$$\begin{aligned} S(t) &\le \frac{\mu}{\alpha} + e^{-\alpha t}\left(S(0) - \frac{\mu}{\alpha}\right),\ S(t) + I(t) \\ &\le \frac{\mu}{\alpha\wedge\beta} + e^{-(\alpha\wedge\beta)t}\left(S(0) + I(0) - \frac{\mu}{\alpha\wedge\beta}\right),\ Z(t) \ge \frac{\eta}{\delta} + e^{-\delta t}\left(Z(0) - \frac{\eta}{\delta}\right). \end{aligned}$$

Accordingly, if $S(0) \le \mu/\alpha$, $S(0)+I(0) \le \mu/(\alpha\wedge\beta)$, $Z(0) \ge \eta/\delta$, then $S(t) \le \mu/\alpha$, $S(t)+I(t) \le \mu/(\alpha\wedge\beta)$ and $Z(t) \ge \eta/\delta$ a.s. Similarly, if $V(0) \le m\mu/(\gamma(\alpha\wedge\beta))$, $I(0)+V(0)+(\nu/\theta)Z(0) \le (\nu\eta(\alpha\wedge\beta)+\mu(m\theta+k\nu))/(\theta(\alpha\wedge\beta)(\beta\wedge\gamma\wedge\delta))$, then $V(t) \le m\mu/(\gamma(\alpha\wedge\beta))$, $I(t)+V(t)+(\nu/\theta)Z(t) \le (\nu\eta(\alpha\wedge\beta)+\mu(m\theta+k\nu))/(\theta(\alpha\wedge\beta)(\beta\wedge\gamma\wedge\delta))$ a.s. In this case, the set

$$\Xi = \Big\{(S, I, V, Z, x)^T \in \mathbb{R}_+^4 \times \mathbb{R} : S \le \frac{\mu}{\alpha}, S + I \le \frac{\mu}{\alpha\wedge\beta}, \\ V \le \frac{m\mu}{\gamma(\alpha\wedge\beta)}, \frac{\eta}{\delta} \le Z, I + V + \frac{\nu}{\theta}Z \le \frac{\nu\eta(\alpha\wedge\beta)+\mu(m\theta+k\nu)}{\theta(\alpha\wedge\beta)(\beta\wedge\gamma\wedge\delta)}\Big\}$$

is positively invariant for the system (1.4).

As a consequence, from now on, we always suppose that the initial value $(S(0), I(0), V(0), Z(0), x(0))^T$ of system (1.4) belongs to the domain Ξ. This shows that the unique global solution $(S(t), I(t), V(t), Z(t), x(t))^T$ of system (1.4) will also belong to the domain Ξ a.s..

3 Existence of a Stationary Distribution

In this section, we will present sufficient criteria for the existence of a stationary distribution which means the coexistence of susceptible monocytes, infected monocytes, free virus particles and immune cells. At first, we present some theories on the existence of a stationary distribution (see Du and Yin 2016).

For a homogeneous Markov process defined in $\mathbb{R}^d$ which is expressed by the stochastic differential equation:

$$\mathrm{d}X(t) = f(X(t))\mathrm{d}t + g(X(t))\mathrm{d}B(t), \tag{3.1}$$

with the initial value $X(0) \in \mathbb{R}^d$, where $B(t)$ is a d-dimensional Brownian motion defined on the complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\ge 0}, \mathbb{P})$. Additionally, $f : \mathbb{R}^d \to \mathbb{R}^d$ and $g : \mathbb{R}^d \to \mathbb{R}^{d\times m}$ are Borel measurable. The following lemma is about the existence of a stationary distribution of system (1.4).

Lemma 3.1 (See Theorem 2.2 in Du and Yin 2016) *Suppose that there is a bounded closed domain $\mathbb{A} \subset \mathbb{R}^d$ with a regular boundary Γ, for any initial value $X(0) \in \mathbb{R}^d$, if*

$$\liminf_{t\to\infty} \frac{1}{t}\int_0^t \mathbb{P}(s, X(s), \mathbb{A})\mathrm{d}s > 0 \text{ a.s.},$$

where $\mathbb{P}(s, X(s), \cdot)$ is the transition probability of $X(t)$. Then system (3.1) will possess a solution which has the Feller property. Besides, system (3.1) admits at least one invariant probability measure $\pi(\cdot)$ on $\mathbb{R}^d$, which means system (3.1) has at least one stationary distribution $\pi(\cdot)$ on $\mathbb{R}^d$.

Lemma 3.2 (Shi and Jiang 2023) *For a stochastic differential equation:*

$$\mathrm{d}(\ln b(t)) = \rho(\ln \bar{b} - \ln b(t)) + \sigma \mathrm{d}B(t),$$

where $\ln \bar{b}$ *and* σ *are positive constants and* $B(t)$ *is a standard Brownian motion. Then for any* $n > 0$,

$$\lim_{t\to\infty} \frac{1}{t}\int_0^t b^n(s)\mathrm{d}s = \bar{b}^n e^{\frac{n^2\sigma^2}{4\rho}}.$$

Theorem 3.1 *Assume that*

$$\mathcal{R}_0^S = \frac{m\bar{b}\mu\delta e^{\frac{\sigma^2}{12\rho}}}{(\alpha\gamma + \bar{b}\mu e^{\frac{\sigma^2}{4\rho}})(\beta\delta + \nu\eta)} > 1,$$

then the stochastic system (1.4) *has at least one stationary distribution* $\pi(\cdot)$ *on* Ξ.

Proof The proof process is divided into three steps: the first two steps are to construct a nonnegative C^2-function $\mathcal{V}(S, I, V, Z, x)$ and a compact set $U \subset \Xi$ such that $L\mathcal{V} \le -1$ for any $(S, I, V, Z, x)^T \in \Xi \setminus U$, the last step is to show the existence of a stationary distribution of system (1.4) by employing Theorem 2.2 in Du and Yin (2016).

Step 1. (Construction of a nonnegative C^2-function and a compact set): For any small enough number $\epsilon \in (0, 1)$, let us define

$$\mathcal{R}_0^S(\epsilon) = \frac{m\bar{b}\mu\delta e^{\frac{\sigma^2}{12\rho}}}{(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)}\bar{b}^2 e^{\frac{\sigma^2}{\rho}})(\gamma + \frac{\bar{b}\mu}{\alpha}e^{\frac{\sigma^2}{4\rho}})(\beta\delta + \nu\eta)}.$$

It is noticeable that $\lim_{\epsilon\to 0^+}\mathcal{R}_0^S(\epsilon) = \mathcal{R}_0^S$. Due to the continuity of the function $\mathcal{R}_0^S(\epsilon)$ with respect to ϵ and $\mathcal{R}_0^S > 1$, we can select ϵ small enough such that $\mathcal{R}_0^S(\epsilon) > 1$.

By system (1.4) and Itô's formula (Mao 1997), one obtains

$$\begin{aligned} L(-\ln S) &= -\frac{\mu}{S} + \alpha + bV \\ &\le -\frac{\mu}{S} + \alpha + \left(\epsilon b^2 + \frac{1}{4\epsilon}\right)V \\ &\le -\frac{\mu}{S} + \alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)}b^2 + \frac{1}{4\epsilon}V, \end{aligned} \tag{3.2}$$

$$L(-\ln I) = -\frac{e^x SV}{I} + \beta + \nu Z, \tag{3.3}$$

$$\begin{aligned} L(-\ln V) &= -\frac{mI}{V} + \gamma + e^x S \\ &\le -\frac{mI}{V} + \gamma + \frac{\mu}{\alpha}e^x, \end{aligned} \tag{3.4}$$

$$\begin{aligned} L(V) &= mI - \gamma V - e^x SV \\ &\leq mI - \gamma V, \end{aligned} \tag{3.5}$$

and

$$L(Z) = \eta + kI + \theta IZ - \delta Z, \tag{3.6}$$

where in the first inequality of (3.2), we have employed the Young inequality

$$b \leq \epsilon b^2 + \frac{1}{4\epsilon},$$

and $\epsilon > 0$ is a small enough number. Define a function $\mathcal{V}_1$ which takes the form

$$\mathcal{V}_1(S, I, V, Z) = -\ln I - c_1 \ln S - c_2 \ln V + \frac{\nu}{\delta} Z + \frac{c_1}{4\gamma\epsilon} V,$$

where c_1 and c_2 are positive constants to be chosen later. In light of (3.2), (3.3), (3.4), (3.5) and (3.6), one has

$$\begin{aligned} L\mathcal{V}_1 \leq & -\frac{e^x SV}{I} - \frac{c_1\mu}{S} - \frac{c_2 mI}{V} + \beta + \nu Z + c_1\alpha + \frac{c_1 m\mu\epsilon}{\gamma(\alpha\wedge\beta)} b^2 \\ & + c_2\gamma + \frac{c_2\mu}{\alpha} e^x + \frac{\nu\eta}{\delta} + \frac{k\nu}{\delta} I + \frac{\theta\nu}{\delta} IZ - \nu Z + \frac{c_1}{4\epsilon} V \\ & + \frac{mc_1}{4\gamma\epsilon} I - \frac{c_1}{4\epsilon} V \\ \leq & -3\sqrt[3]{c_1 c_2 m\mu e^x} + \beta + \frac{\nu\eta}{\delta} + c_1\left(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)} \bar{b}^2 e^{\frac{\sigma^2}{\rho}}\right) \\ & + c_2\left(\gamma + \frac{\bar{b}\mu}{\alpha} e^{\frac{\sigma^2}{4\rho}}\right) + \left(\frac{k\nu}{\delta} + \frac{mc_1}{4\gamma\epsilon}\right) I + \frac{\theta\nu}{\delta} IZ \\ & + \frac{c_1 m\mu\epsilon}{\gamma(\alpha\wedge\beta)} (b^2 - \bar{b}^2 e^{\frac{\sigma^2}{\rho}}) + \frac{c_2\mu}{\alpha}(b - \bar{b} e^{\frac{\sigma^2}{4\rho}}) \\ = & -3\sqrt[3]{c_1 c_2 m\mu e^{\tilde{x}}} + \beta + \frac{\nu\eta}{\delta} + c_1\left(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)} \bar{b}^2 e^{\frac{\sigma^2}{\rho}}\right) \\ & + c_2\left(\gamma + \frac{\bar{b}\mu}{\alpha} e^{\frac{\sigma^2}{4\rho}}\right) + \left(\frac{k\nu}{\delta} + \frac{mc_1}{4\gamma\epsilon}\right) I + \frac{\theta\nu}{\delta} IZ \\ & + 3(\sqrt[3]{c_1 c_2 m\mu e^{\tilde{x}}} - \sqrt[3]{c_1 c_2 m\mu e^x}) + \frac{c_1 m\mu\epsilon}{\gamma(\alpha\wedge\beta)} (b^2 - \bar{b}^2 e^{\frac{\sigma^2}{\rho}}) + \frac{c_2\mu}{\alpha}(b - \bar{b} e^{\frac{\sigma^2}{4\rho}}). \end{aligned}$$

For the equation

$$\mathrm{d}x = \rho(\bar{x} - x)\mathrm{d}t + \sigma \mathrm{d}B(t).$$

By using the references (Cai et al. 2018; Lin et al. 2014; Zhang and Yuan 2021), one can get that the Markov process x has the ergodic property and it will weakly converge

to an invariant density

$$\pi(y) = \frac{\sqrt{\rho}}{\sqrt{\pi}\sigma} e^{-\frac{\rho(y-\bar{x})^2}{\sigma^2}}, \ y \in \mathbb{R},$$

which together with the ergodic theorem (Kutoyants 2003), one obtains

$$\lim_{t\to\infty} \frac{1}{t} \int_0^t e^{\frac{x(s)}{3}} \mathrm{d}s = \int_{-\infty}^{\infty} e^{\frac{y}{3}} \pi(y) \mathrm{d}y = e^{\frac{\bar{x}}{3} + \frac{\sigma^2}{36\rho}},$$

and hence

$$e^{\tilde{x}} = \left(\int_{-\infty}^{\infty} e^{\frac{y}{3}} \pi(y) \mathrm{d}y \right)^3 = \bar{b} e^{\frac{\sigma^2}{12\rho}}.$$

Choose

$$c_1 = \frac{\bar{b} m \mu e^{\frac{\sigma^2}{12\rho}}}{(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)} \bar{b}^2 e^{\frac{\sigma^2}{\rho}})^2 (\gamma + \frac{\bar{b}\mu}{\alpha} e^{\frac{\sigma^2}{4\rho}})}, \ c_2 = \frac{\bar{b} m \mu e^{\frac{\sigma^2}{12\rho}}}{(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)} \bar{b}^2 e^{\frac{\sigma^2}{\rho}})(\gamma + \frac{\bar{b}\mu}{\alpha} e^{\frac{\sigma^2}{4\rho}})^2},$$

and then

$$\begin{aligned}
L\mathcal{V}_1 \le & -\frac{\bar{b} m \mu e^{\frac{\sigma^2}{12\rho}}}{(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)} \bar{b}^2 e^{\frac{\sigma^2}{\rho}})(\gamma + \frac{\bar{b}\mu}{\alpha} e^{\frac{\sigma^2}{4\rho}})} + \beta + \frac{\nu\eta}{\delta} + \left(\frac{k\nu}{\delta} + \frac{mc_1}{4\gamma\epsilon} \right) I \\
& + \frac{\theta^2[\nu\eta(\alpha\wedge\beta) + \mu(m\theta + k\nu)]}{\delta\theta(\alpha\wedge\beta)(\beta\wedge\gamma\wedge\delta)} I \\
& + 3(\sqrt[3]{c_1 c_2 m \mu e^{\tilde{x}}} - \sqrt[3]{c_1 c_2 m \mu e^{x}}) + \frac{c_1 m \mu \epsilon}{\gamma(\alpha\wedge\beta)} (b^2 - \bar{b}^2 e^{\frac{\sigma^2}{\rho}}) + \frac{c_2 \mu}{\alpha} (b - \bar{b} e^{\frac{\sigma^2}{4\rho}}) \\
:= & -\frac{\beta\delta + \nu\eta}{\delta} (\mathcal{R}_0^S(\epsilon) - 1) + K_6 I + 3(\sqrt[3]{c_1 c_2 m \mu e^{\tilde{x}}} - \sqrt[3]{c_1 c_2 m \mu e^{x}}) \\
& + \frac{c_1 m \mu \epsilon}{\gamma(\alpha\wedge\beta)} (b^2 - \bar{b}^2 e^{\frac{\sigma^2}{\rho}}) + \frac{c_2 \mu}{\alpha} (b - \bar{b} e^{\frac{\sigma^2}{4\rho}}),
\end{aligned} \tag{3.7}$$

where

$$\mathcal{R}_0^S(\epsilon) = \frac{m \bar{b} \mu \delta e^{\frac{\sigma^2}{12\rho}}}{(\alpha + \frac{m\mu\epsilon}{\gamma(\alpha\wedge\beta)} \bar{b}^2 e^{\frac{\sigma^2}{\rho}})(\gamma + \frac{\bar{b}\mu}{\alpha} e^{\frac{\sigma^2}{4\rho}})(\beta\delta + \nu\eta)}$$

and

$$K_6 = \frac{k\nu}{\delta} + \frac{mc_1}{4\gamma\epsilon} + \frac{\theta^2[\nu\eta(\alpha\wedge\beta) + \mu(m\theta + k\nu)]}{\delta\theta(\alpha\wedge\beta)(\beta\wedge\gamma\wedge\delta)}.$$

Next, let us define

$$\mathcal{V}_2(S) = -\ln S,\ \mathcal{V}_3(V) = -\ln V,\ \mathcal{V}_4(Z) = -\ln Z,\ \mathcal{V}_5(x) = e^x - 1 - x,$$

then applying Itô's formula (Mao 1997) to $\mathcal{V}_2$, $\mathcal{V}_3$, $\mathcal{V}_4$ and $\mathcal{V}_5$, respectively, we have

$$\begin{aligned} L\mathcal{V}_2 &= -\frac{\mu}{S} + \alpha + e^x V \\ &\le -\frac{\mu}{S} + \alpha + \frac{m\mu}{\gamma(\alpha \wedge \beta)} e^x, \end{aligned} \tag{3.8}$$

$$\begin{aligned} L\mathcal{V}_3 &= -\frac{mI}{V} + \gamma + e^x S \\ &\le -\frac{mI}{V} + \gamma + \frac{\mu}{\alpha} e^x, \end{aligned} \tag{3.9}$$

$$\begin{aligned} L\mathcal{V}_4 &= -\frac{\eta}{Z} - \frac{kI}{Z} - \theta I + \delta \\ &\le -\frac{\eta}{Z} + \delta, \end{aligned} \tag{3.10}$$

and

$$L\mathcal{V}_5 = \rho\bar{x}e^x - \rho\bar{x} + \rho x - \rho x e^x + \frac{\sigma^2}{2} e^x. \tag{3.11}$$

Define a C^2-function $\bar{\mathcal{V}}(S, I, V, Z, x) : \Xi \to \mathbb{R}$ as follows

$$\bar{\mathcal{V}}(S, I, V, Z, x) = M\mathcal{V}_1(S, I, V, Z) + \mathcal{V}_2(S) + \mathcal{V}_3(V) + \mathcal{V}_4(Z) + \mathcal{V}_5(x),$$

where $M > 0$ is a sufficiently large constant satisfying the condition

$$-\frac{M\beta\delta + \nu\eta}{\delta}(\mathcal{R}_0^S(\epsilon) - 1) + K_7 \le -2, \tag{3.12}$$

and

$$K_7 := \sup_{x\in\mathbb{R}} \left\{ \rho\bar{x}e^x + \rho x - \rho x e^x + \left(\frac{m\mu}{\gamma(\alpha \wedge \beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \right\} < \infty.$$

On the other hand, note that $\bar{\mathcal{V}}(S, I, V, Z, x)$ is not only continuous, but also tends to ∞ as $(S, I, V, Z, x)^T$ approaches the boundary of Ξ. Accordingly, it has a lower bound and achieves this lower bound at a point $(S_0, I_0, V_0, Z_0, x_0)^T$ in the interior of Ξ. Then we define a C^2-function $\mathcal{V} : \Xi \to \overline{\mathbb{R}}_+$ which takes the form

$$\begin{aligned} \mathcal{V}(S, I, V, Z, x) &= \bar{\mathcal{V}}(S, I, V, Z, x) - \bar{\mathcal{V}}(S_0, I_0, V_0, Z_0, x_0) \\ &= M\mathcal{V}_1(S, I, V, Z) + \mathcal{V}_2(S) + \mathcal{V}_3(V) + \mathcal{V}_4(Z) \\ &\quad + \mathcal{V}_5(x) - \bar{\mathcal{V}}(S_0, I_0, V_0, Z_0, x_0). \end{aligned}$$

According to (3.7), (3.8), (3.9), (3.10) and (3.11), one has

$$\begin{aligned}
L\mathcal{V} \leq & -\frac{M\beta\delta+\nu\eta}{\delta}(\mathcal{R}_0^S(\epsilon)-1)+MK_6I-\frac{\mu}{S}-\frac{mI}{V}-\frac{\eta}{Z} \\
& +\rho\bar{x}e^x+\rho x-\rho xe^x+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x+\alpha \\
& +\gamma+\delta-\rho\bar{x}+3M(\sqrt[3]{c_1c_2m\mu e^{\tilde{x}}}-\sqrt[3]{c_1c_2m\mu e^x}) \\
& +\frac{Mc_1m\mu\epsilon}{\gamma(\alpha\wedge\beta)}(b^2-\bar{b}^2e^{\frac{\sigma^2}{\rho}})+\frac{Mc_2\mu}{\alpha}(b-\bar{b}e^{\frac{\sigma^2}{4\rho}}) \\
:= & G(S,I,V,Z,x)+3M(\sqrt[3]{c_1c_2m\mu e^{\tilde{x}}}-\sqrt[3]{c_1c_2m\mu e^x}) \\
& +\frac{Mc_1m\mu\epsilon}{\gamma(\alpha\wedge\beta)}(b^2-\bar{b}^2e^{\frac{\sigma^2}{\rho}})+\frac{Mc_2\mu}{\alpha}(b-\bar{b}e^{\frac{\sigma^2}{4\rho}}),
\end{aligned} \tag{3.13}$$

where

$$\begin{aligned}
G(S,I,V,Z,x) = & -\frac{M\beta\delta+\nu\eta}{\delta}(\mathcal{R}_0^S(\epsilon)-1)+MK_6I \\
& -\frac{\mu}{S}-\frac{mI}{V}-\frac{\eta}{Z}+\rho\bar{x}e^x+\rho x-\rho xe^x+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x \\
& +\alpha+\gamma+\delta-\rho\bar{x}.
\end{aligned}$$

Step 2. (Construction of a compact set): Now we are in the position to construct a compact set U_ϵ which takes the form

$$U_\epsilon=\left\{(S,I,V,Z,x)^T\in\Xi: S\geq\epsilon, I\geq\epsilon, V\geq\epsilon^2, Z\geq\epsilon, \epsilon\leq e^x\leq\frac{1}{\epsilon}\right\},$$

where $\epsilon\in(0,1)$ is a small enough number which ensures the following inequalities hold

$$-\frac{\mu}{\epsilon}+K_8\leq-1, \tag{3.14}$$

$$\epsilon\leq\frac{1}{MK_6}, \tag{3.15}$$

$$-\frac{m}{\epsilon}+K_8\leq-1, \tag{3.16}$$

$$-\frac{\eta}{\epsilon}+K_8\leq-1, \tag{3.17}$$

$$-\frac{\rho\ln\frac{1}{\epsilon}}{2\epsilon}+K_9\leq-1, \tag{3.18}$$

$$\frac{\rho\ln\epsilon}{2}+K_{10}\leq-1. \tag{3.19}$$

Here K_8, K_9 and K_{10} are positive constants explicitly presented in the expressions (3.21), (3.26) and (3.28), respectively. Next, we can divide the set $\Xi\backslash U_\epsilon$ into six subsets $U^c_{\epsilon,i}$, $i = 1, \ldots, 6$, where

$$
\begin{aligned}
&U^c_{\epsilon,1} = \{(S, I, V, Z, x)^T \in \Xi : S < \epsilon\}, U^c_{\epsilon,2} = \{(S, I, V, Z, x)^T \in \Xi : I < \epsilon\},\\
&U^c_{\epsilon,3} = \{(S, I, V, Z, x)^T \in \Xi : I \geq \epsilon, V < \epsilon^2\},\\
&U^c_{\epsilon,4} = \{(S, I, V, Z, x)^T \in \Xi : Z < \epsilon\}, U^c_{\epsilon,5} = \left\{(S, I, V, Z, x)^T \in \Xi : x > \ln\frac{1}{\epsilon}\right\},\\
&U^c_{\epsilon,6} = \{(S, I, V, Z, x)^T \in \Xi : x < \ln\epsilon\}.
\end{aligned}
$$

Apparently, $\Xi\backslash U_\epsilon = \bigcup_{i=1}^6 U^c_{\epsilon,i}$. Next, we will verify that $G(S, I, V, Z, x) \leq -1$ for any $(S, I, V, Z, x) \in U^c_\epsilon$. This is equivalent to verifying its satisfaction on the above six subdomains.

Case 1. If $(S, I, V, Z, x)^T \in U^c_{\epsilon,1}$, from (3.13) it follows that

$$
\begin{aligned}
G(S, I, V, Z, x) &\leq -\frac{\mu}{S} + MK_6 I + \rho\bar{x}e^x + \rho x - \rho x e^x\\
&\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2}\right)e^x + \alpha + \gamma + \delta - \rho\bar{x}\\
&\leq -\frac{\mu}{\epsilon} + \frac{MK_6\mu}{\alpha\wedge\beta} + \rho\bar{x}e^x + \rho x - \rho x e^x\\
&\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2}\right)e^x + \alpha + \gamma + \delta - \rho\bar{x}\\
&\leq -\frac{\mu}{\epsilon} + K_8\\
&\leq -1,
\end{aligned}
\tag{3.20}
$$

which follows from (3.14) and

$$
\begin{aligned}
K_8 := \sup_{x\in\mathbb{R}} &\left\{\rho\bar{x}e^x + \rho x - \rho x e^x + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2}\right)e^x\right.\\
&\left. + \frac{MK_6\mu}{\alpha\wedge\beta} + \alpha + \gamma + \delta - \rho\bar{x}\right\} < \infty.
\end{aligned}
\tag{3.21}
$$

Case 2. If $(S, I, V, Z, x)^T \in U^c_{\epsilon,2}$, by means of (3.13), one gets

$$
\begin{aligned}
G(S, I, V, Z, x) &\leq -\tfrac{M\beta\delta+\nu\eta}{\delta}(\mathcal{R}^S_0(\epsilon) - 1) + MK_6 I + \rho\bar{x}e^x + \rho x - \rho x e^x\\
&\quad + \left(\tfrac{m\mu}{\gamma(\alpha\wedge\beta)} + \tfrac{\mu}{\alpha} + \tfrac{\sigma^2}{2}\right)e^x + \alpha + \gamma\\
&\quad + \delta - \rho\bar{x}\\
&\leq -\tfrac{M\beta\delta+\nu\eta}{\delta}(\mathcal{R}^S_0(\epsilon) - 1) + MK_6\epsilon + K_7 \leq -1,
\end{aligned}
\tag{3.22}
$$

which follows from (3.12) and (3.15) and

$$K_7 := \sup_{x\in\mathbb{R}} \left\{ \rho\bar{x}e^x + \rho x - \rho x e^x + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \right\} < \infty.$$

Case 3. If $(S, I, V, Z, x)^T \in U^c_{\epsilon,3}$, according to (3.13), one has

$$\begin{aligned} G(S, I, V, Z, x) &\le -\frac{mI}{V} + MK_6 I + \rho\bar{x}e^x + \rho x - \rho x e^x \\ &\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \\ &\le -\frac{m\epsilon}{\epsilon^2} + \frac{MK_6\mu}{\alpha\wedge\beta} + \rho\bar{x}e^x + \rho x - \rho x e^x \\ &\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \\ &\le -\frac{m}{\epsilon} + K_8 \\ &\le -1, \end{aligned} \tag{3.23}$$

which follows from (3.16).

Case 4. If $(S, I, V, Z, x)^T \in U^c_{\epsilon,4}$, in view of (3.13), one obtains

$$\begin{aligned} G(S, I, V, Z, x) &\le -\frac{\eta}{Z} + MK_6 I + \rho\bar{x}e^x + \rho x - \rho x e^x \\ &\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \\ &\le -\frac{\eta}{\epsilon} + \frac{MK_6\mu}{\alpha\wedge\beta} + \rho\bar{x}e^x + \rho x - \rho x e^x \\ &\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \\ &\le -\frac{\eta}{\epsilon} + K_8 \\ &\le -1, \end{aligned} \tag{3.24}$$

which follows from (3.17).

Case 5. If $(S, I, V, Z, x)^T \in U^c_{\epsilon,5}$, from (3.13) it follows that

$$\begin{aligned} G(S, I, V, Z, x) &\le -\frac{\rho}{2}xe^x + MK_6 I + \rho\bar{x}e^x + \rho x - \frac{\rho}{2}xe^x \\ &\quad + \left(\frac{m\mu}{\gamma(\alpha\wedge\beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2} \right) e^x + \alpha + \gamma + \delta - \rho\bar{x} \\ &\le -\frac{\rho \ln\frac{1}{\epsilon}}{2\epsilon} + \frac{MK_6\mu}{\alpha\wedge\beta} + \rho\bar{x}e^x + \rho x - \frac{\rho}{2}xe^x \end{aligned}$$

$$
\begin{aligned}
&+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x+\alpha+\gamma+\delta-\rho\bar{x}\\
&\leq -\frac{\rho\ln\frac{1}{\epsilon}}{2\epsilon}+K_9\leq -1,
\end{aligned}
\tag{3.25}
$$

which follows from (3.18) and

$$
\begin{aligned}
K_9 := \sup_{x\in\mathbb{R}}\Bigg\{&\rho\bar{x}e^x+\rho x-\frac{\rho}{2}xe^x+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x\\
&+\frac{MK_6\mu}{\alpha\wedge\beta}+\alpha+\gamma+\delta-\rho\bar{x}\Bigg\}<\infty.
\end{aligned}
\tag{3.26}
$$

Case 6. If $(S, I, V, Z, x)^T \in U^c_{\epsilon,6}$, according to (3.13), one gets

$$
\begin{aligned}
G(S,I,V,Z,x)&\leq \frac{\rho}{2}x+MK_6I+\rho\bar{x}e^x+\frac{\rho}{2}x-\rho xe^x\\
&\quad+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x+\alpha+\gamma+\delta-\rho\bar{x}\\
&\leq \frac{\rho\ln\epsilon}{2}+\frac{MK_6\mu}{\alpha\wedge\beta}+\rho\bar{x}e^x+\frac{\rho}{2}x-\rho xe^x\\
&\quad+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x+\alpha+\gamma+\delta-\rho\bar{x}\\
&\leq \frac{\rho\ln\epsilon}{2}+K_{10}\\
&\leq -1,
\end{aligned}
\tag{3.27}
$$

which follows from (3.19) and

$$
\begin{aligned}
K_{10} := \sup_{x\in\mathbb{R}}\Bigg\{&\rho\bar{x}e^x+\frac{\rho}{2}x-\rho xe^x+\left(\frac{m\mu}{\gamma(\alpha\wedge\beta)}+\frac{\mu}{\alpha}+\frac{\sigma^2}{2}\right)e^x\\
&+\frac{MK_6\mu}{\alpha\wedge\beta}+\alpha+\gamma+\delta-\rho\bar{x}\Bigg\}<\infty.
\end{aligned}
\tag{3.28}
$$

On the basis of (3.20), (3.22), (3.23), (3.24), (3.25) and (3.27), it can be concluded that there is a small enough ϵ such that

$$
G(S,I,V,Z,x)\leq -1 \text{ for any } (S,I,V,Z,x)^T\in\Xi\setminus U_\epsilon. \tag{3.29}
$$

Let

$$K_{11} := \sup_{(S,I,V,Z,x)^T \in \Xi} \Big\{ -\frac{M\beta\delta + \nu\eta}{\delta}(\mathcal{R}_0^S(\epsilon) - 1) + MK_6 I - \frac{\mu}{S} - \frac{mI}{V} - \frac{\eta}{Z} + \rho\bar{x}e^x + \rho x - \rho x e^x + \Big(\frac{m\mu}{\gamma(\alpha \wedge \beta)} + \frac{\mu}{\alpha} + \frac{\sigma^2}{2}\Big)e^x + \alpha + \gamma + \delta - \rho\bar{x} \Big\}.$$

Then

$$G(S, I, V, Z, x) \le K_{11} < \infty \text{ for any } (S, I, V, Z, x)^T \in \Xi. \tag{3.30}$$

Step 3. (Existence of a stationary distribution): For any initial value $(S(0), I(0), V(0), Z(0), x(0))^T \in \Xi$, integrating both sides of (3.13) from 0 to t and then taking the expectation give that

$$\begin{aligned} 0 &\le \frac{\mathbb{E}\mathcal{V}(S(t), I(t), V(t), Z(t), x(t))}{t} \\ &= \frac{\mathbb{E}\mathcal{V}(S(0), I(0), V(0), Z(0), x(0))}{t} + \frac{1}{t}\int_0^t \mathbb{E}(L\mathcal{V}(S(s), I(s), V(s), Z(s), x(s)))\mathrm{d}s \\ &\le \frac{\mathbb{E}\mathcal{V}(S(0), I(0), V(0), Z(0), x(0))}{t} + \frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathrm{d}s \\ &\quad + 3M\sqrt[3]{c_1c_2m\mu}\mathbb{E}\Big(\int_{-\infty}^{\infty} e^{\frac{y}{3}}\pi(y)dy - \frac{1}{t}\int_0^t e^{\frac{x(s)}{3}}\mathrm{d}s\Big) \\ &\quad + \frac{Mc_1m\mu\epsilon}{\gamma(\alpha \wedge \beta)}\mathbb{E}\Big(\frac{1}{t}\int_0^t b^2(s)\mathrm{d}s - \bar{b}^2 e^{\frac{\sigma^2}{\rho}}\Big) \\ &\quad + \frac{Mc_2\mu}{\alpha}\mathbb{E}\Big(\frac{1}{t}\int_0^t b(s)\mathrm{d}s - \bar{b}e^{\frac{\sigma^2}{4\rho}}\Big). \end{aligned} \tag{3.31}$$

On the basis of the ergodicity of $x(t)$ and the strong law of large numbers (Lipster 1980), one can see that

$$\lim_{t\to\infty} \mathbb{E}\Big(\int_{-\infty}^{\infty} e^{\frac{y}{3}}\pi(y)dy - \frac{1}{t}\int_0^t e^{\frac{x(s)}{3}}\mathrm{d}s\Big) = \int_{-\infty}^{\infty} e^{\frac{y}{3}}\pi(y)\mathrm{d}y - \int_{-\infty}^{\infty} e^{\frac{y}{3}}\pi(y)\mathrm{d}y = 0 \text{ a.s.}, \tag{3.32}$$

$$\lim_{t\to\infty} \mathbb{E}\Big(\frac{1}{t}\int_0^t b^2(s)\mathrm{d}s - \bar{b}^2 e^{\frac{\sigma^2}{\rho}}\Big) = \bar{b}^2 e^{\frac{\sigma^2}{\rho}} - \bar{b}^2 e^{\frac{\sigma^2}{\rho}} = 0 \text{ a.s.}, \tag{3.33}$$

and

$$\lim_{t\to\infty} \mathbb{E}\Big(\frac{1}{t}\int_0^t b(s)\mathrm{d}s - \bar{b}e^{\frac{\sigma^2}{4\rho}}\Big) = \bar{b}e^{\frac{\sigma^2}{4\rho}} - \bar{b}e^{\frac{\sigma^2}{4\rho}} = 0 \text{ a.s.} \tag{3.34}$$

Taking the inferior limit on both sides of (3.31) and combining with (3.32), (3.33) and (3.34), one gets

$$\begin{aligned} 0 &\leq \liminf_{t\to\infty} \frac{\mathbb{E}\mathcal{V}(S(0), I(0), V(0), Z(0), x(0))}{t} \\ &\quad + \liminf_{t\to\infty} \frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathrm{d}s \\ &= \liminf_{t\to\infty} \frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathrm{d}s \text{ a.s.} \end{aligned} \tag{3.35}$$

On the other hand, from (3.29) and (3.30) it follows that

$$\begin{aligned} &\frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathrm{d}s \\ &\quad = \frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathbf{1}_{\{(S(s),I(s),V(s),Z(s),x(s))^T\in U_\epsilon\}}\mathrm{d}s \\ &\qquad + \frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathbf{1}_{\{(S(s),I(s),V(s),Z(s),x(s))^T\in(\Xi\setminus U_\epsilon)\}}\mathrm{d}s \\ &\quad \leq \frac{K_{11}}{t}\int_0^t \mathbf{1}_{\{(S(s),I(s),V(s),Z(s),x(s))^T\in U_\epsilon\}}\mathrm{d}s - \frac{1}{t}\int_0^t \mathbf{1}_{\{(S(s),I(s),V(s),Z(s),x(s))^T\in(\Xi\setminus U_\epsilon)\}}\mathrm{d}s \\ &\quad \leq \frac{K_{11}+1}{t}\int_0^t \mathbf{1}_{\{(S(s),I(s),V(s),Z(s),x(s))^T\in U_\epsilon\}}\mathrm{d}s - 1. \end{aligned} \tag{3.36}$$

By (3.35) and (3.36), it is inferred that

$$\begin{aligned} &\liminf_{t\to\infty} \frac{1}{t}\int_0^t \mathbf{1}_{\{(S(s),I(s),V(s),Z(s),x(s))^T\in U_\epsilon\}}\mathrm{d}s \\ &\quad \geq \liminf_{t\to\infty} \frac{1}{K_{11}+1}\left(1 + \frac{1}{t}\int_0^t \mathbb{E}(G(S(s), I(s), V(s), Z(s), x(s)))\mathrm{d}s\right) \\ &\quad \geq \frac{1}{K_{11}+1} > 0 \text{ a.s.} \end{aligned} \tag{3.37}$$

Let $\mathbb{P}(t, (S, I, V, Z, x), \mathbb{A})$ be the transition probability of $(S(t), I(t), V(t), Z(t), x(t))^T$ belonging to the set $\mathbb{A}$. Applying Fatou's lemma (Du and Yin 2016) to (3.37) yields that

$$\liminf_{t\to\infty} \frac{1}{t}\int_0^t \mathbb{P}(s, (S(s), I(s), V(s), Z(s), x(s)), U_\epsilon)\mathrm{d}s \geq \frac{1}{K_{11}+1} > 0 \text{ a.s.}$$

Consequently, in light of Lemma 3.1, we can get that system (1.4) admits at least one stationary distribution $\pi(\cdot)$ on Ξ, which has the Feller property. The proof is confirmed. □

4 Probability Density of System (1.4) Around E^*

In this section, we will concentrate on obtaining the specific form of covariance matrix in the probability density around the quasi-endemic equilibrium of system (1.4). To begin with, we need to present an equivalent transformation of system (1.4).

4.1 Equivalent Transformation of System (1.4)

To start with, we define a quasi-endemic equilibrium $E^* = (S^*, I^*, V^*, Z^*, x^*)^T$ involved in stochasticity by the equations

$$\begin{cases} \mu - \alpha S^* - e^{x^*} S^* V^* = 0, \\ e^{x^*} S^* V^* - \beta I^* - \nu I^* Z^* = 0, \\ m I^* - \gamma V^* - e^{x^*} S^* V^* = 0, \\ \eta + k I^* + \theta I^* Z^* - \delta Z^* = 0, \\ \rho(\bar{x} - x^*) = 0. \end{cases} \tag{4.1}$$

By solving Eq. (4.1), one can see that if

$$\mathcal{R}_0^P = \sqrt{\frac{m\bar{b}\mu\delta}{(\alpha\gamma + \bar{b}\mu)(\beta\delta + \nu\eta)}} > 1,$$

then

$$S^* = S^+ > 0,\ I^* = I^+ > 0,\ V^* = V^+ > 0,\ Z^* = Z^+ > 0,\ x^* = \bar{x} > 0,$$

where S^+, I^+, V^+ and Z^+ are the same as in Section one.

Let $X = (u_1, u_2, u_3, u_4, u_5)^T = (S - S^*, I - I^*, V - V^*, Z - Z^*, x - \bar{x})^T$. Based on Itô's integral and system (1.4), the corresponding linearized system of (1.4) near E^* takes the form

$$\begin{cases} \mathrm{d}u_1 = (-a_{11}u_1 - a_{13}u_3 - a_{15}u_5)\mathrm{d}t, \\ \mathrm{d}u_2 = (a_{21}u_1 - a_{22}u_2 + a_{23}u_3 - a_{24}u_4 + a_{25}u_5)\mathrm{d}t, \\ \mathrm{d}u_3 = (-a_{31}u_1 + a_{32}u_2 - a_{33}u_3 - a_{35}u_5)\mathrm{d}t, \\ \mathrm{d}u_4 = (a_{42}u_2 - a_{44}u_4)\mathrm{d}t, \\ \mathrm{d}u_5 = -\rho u_5 \mathrm{d}t + \sigma \mathrm{d}B(t), \end{cases} \tag{4.2}$$

where

$$\begin{aligned} &a_{11} = \alpha + e^{x^*} V^* > 0,\ a_{13} = e^{x^*} S^* > 0,\ a_{15} = e^{x^*} S^* V^* > 0, \\ &a_{21} = e^{x^*} V^* > 0,\ a_{22} = \beta + \nu Z^* > 0,\ a_{23} = e^{x^*} S^* > 0, \\ &a_{24} = \nu I^* > 0,\ a_{25} = e^{x^*} S^* V^* > 0,\ a_{31} = e^{x^*} V^* > 0,\ a_{32} = m > 0, \\ &a_{33} = \gamma + e^{x^*} S^* > 0,\ a_{35} = e^{x^*} S^* V^* > 0, \\ &a_{42} = k + \theta Z^* > 0,\ a_{44} = \delta - \theta I^* = \frac{\eta + k I^*}{Z^*} > 0. \end{aligned}$$

Apparently, $a_{13} = a_{23}, a_{21} = a_{31}, a_{15} = a_{25} = a_{35}$.

Before giving the specific form of covariance matrix in the probability density, we should give a definition and a lemma.

Definition 4.1 (Ma et al. 2015) The characteristic polynomial of the square matrix A_n is defined as $\varphi_{A_n}(\lambda) = \lambda^n + a_1\lambda^{n-1} + \ldots + a_{n-1}\lambda + a_n$, then A_n is called a Hurwitz matrix if and only if A_n has all negative real-part eigenvalues, i.e.,

$$H_k = \begin{vmatrix} a_1 & a_3 & a_5 & \ldots & a_{2k-1} \\ 1 & a_2 & a_4 & \ldots & a_{2k-2} \\ 0 & a_1 & a_3 & \ldots & a_{2k-3} \\ 0 & 1 & a_2 & \ldots & a_{2k-4} \\ \ldots & \ldots & \ldots & \ldots & \ldots \\ 0 & 0 & 0 & \ldots & a_k \end{vmatrix} > 0, \ k = 1, \ldots, n,$$

where the complementary definition is $a_j = 0, j > n$. Furthermore, the corresponding necessary conditions for A_n to be a Hurwitz matrix are given as follows

$$(i)\ a_j > 0, \ j = 1, \ldots, n; \ (ii)\ a_i a_{i+1} > a_{i-1}a_{i+2}, \ i = 1, \ldots, n-2, \ a_0 = 1.$$

Lemma 4.1 (Zhou et al. 2023) *For any real matrices $A = (a_{ij})_{5\times 5}$ and $M_c = diag\{\rho_1, \rho_2, \rho_3, \rho_4, \rho_5\}$, where $\rho_k > 0$ $(\forall k = 1, 2, \ldots, 5)$, let $\Sigma_0 = (\vartheta_{ij})_{5\times 5}$ be a symmetric matrix, which satisfies the five-dimensional algebraic equation*

$$M_c + A\Sigma_0 + \Sigma_0 A^T = 0.$$

If A is an n-dimensional Hurwitz matrix, then Σ_0 is positive definite.

Remark 4.1 Since the proof of this lemma is very complicated, the uniqueness and exact form of Σ_0 can be obtained in the proof process.

For the sake of simplicity, denote by

$$C = \begin{pmatrix} -a_{11} & 0 & -a_{13} & 0 & -a_{15} \\ a_{21} & -a_{22} & a_{23} & -a_{24} & a_{25} \\ -a_{31} & a_{32} & -a_{33} & 0 & -a_{35} \\ 0 & a_{42} & 0 & -a_{44} & 0 \\ 0 & 0 & 0 & 0 & -\rho \end{pmatrix}, \ H = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma \end{pmatrix}.$$

With these notations, system (4.2) can be written in the following form:

$$\mathrm{d}X(t) = CX(t)\mathrm{d}t + H\mathrm{d}B(t). \tag{4.3}$$

Theorem 4.1 *If $\mathcal{R}_0^P > 1$, then the stationary solution $(S(t), I(t), V(t), Z(t), x(t))^T$ of system (1.4) near E^* follows a normal distribution $\mathbb{N}(E^*, \Sigma)$, where the specific form of covariance matrix Σ can be obtained in the proof process.*

Proof By the monographs of Mao (1997) and Oksendal (2000), one can see that for any initial value $X(0) = (u_1(0), u_2(0), u_3(0), u_4(0), u_5(0))^T$, system (4.3) has a unique exact solution

$$X(t) = e^{Ct}X(0) + \int_0^t e^{C(t-s)}H\mathrm{d}B(s).$$

Since the martingale $\int_0^t e^{C(t-s)}H\mathrm{d}B(s)$ follows the normal distribution $\mathbb{N}(0, \bar{\Sigma}(t))$ at time t, where

$$\bar{\Sigma}(t) = \int_0^t e^{C(t-s)H^2}e^{C^T(t-s)}\mathrm{d}s.$$

Then $X(t)$ follows a unique normal distribution $\mathbb{N}(e^{Ct}X(0), \bar{\Sigma}(t))$ at time t.

On the other hand, denote by

$$C^{(4)} = \begin{pmatrix} -a_{11} & 0 & -a_{13} & 0 \\ a_{21} & -a_{22} & a_{23} & -a_{24} \\ -a_{31} & a_{32} & -a_{33} & 0 \\ 0 & a_{42} & 0 & -a_{44} \end{pmatrix},$$

where $C^{(4)}$ is a four-order real symmetric matrix of the matrix C. Since the statistical properties of S, I, V, Z in the system (1.4) are what we need to consider, then we will illustrate that all the eigenvalues of $C^{(4)}$ have negative real-parts. For this purpose, let us define the characteristic equation of the matrix $C^{(4)}$ by

$$\varphi_{C^{(4)}}(\lambda) = \lambda^4 + \tau_1\lambda^3 + \tau_2\lambda^2 + \tau_3\lambda + \tau_4 = 0,$$

where

$$\begin{aligned}
\tau_1 &= a_{11} + a_{22} + a_{33} + a_{44} > 0, \\
\tau_2 &= a_{11}a_{22} + a_{11}a_{33} + a_{11}a_{44} + a_{22}a_{33} + a_{22}a_{44} + a_{33}a_{44} \\
&\quad + a_{24}a_{42} - a_{13}a_{31} - a_{23}a_{32} > 0, \\
\tau_3 &= a_{11}a_{22}a_{33} + a_{11}a_{22}a_{44} + a_{11}a_{33}a_{44} + a_{11}a_{24}a_{42} + a_{13}a_{21}a_{32} \\
&\quad + a_{22}a_{33}a_{44} + a_{24}a_{33}a_{42} - a_{11}a_{23}a_{32} - a_{13}a_{22}a_{31} - a_{13}a_{31}a_{44} - a_{23}a_{32}a_{44} > 0, \\
\tau_4 &= a_{11}a_{22}a_{33}a_{44} + a_{11}a_{24}a_{33}a_{42} \\
&\quad + a_{13}a_{21}a_{32}a_{44} - a_{11}a_{23}a_{32}a_{44} - a_{13}a_{22}a_{31}a_{44} - a_{13}a_{24}a_{31}a_{42} > 0.
\end{aligned}$$

According to simple computation, one can see that

$$\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4 > 0.$$

On the basis of the above results and Definition 4.1, one can get that the matrix $C^{(4)}$ has all negative real-part eigenvalues. Since the characteristic equation of the matrix

C is

$$\varphi_C(\lambda) = (\lambda + \rho)(\lambda^4 + \tau_1\lambda^3 + \tau_2\lambda^2 + \tau_3\lambda + \tau_4) = 0,$$

as a result, the matrix C also has negative real-part eigenvalues. Thereby, combining the stability theory of zero solution with the general linear equation (Ma et al. 2015), one can obtain that $\lim_{t\to\infty} e^{Ct} = 0$ and

$$\lim_{t\to\infty} e^{Ct}X(0) = 0,$$
$$\Sigma := \lim_{t\to\infty} \bar{\Sigma}(t) = \lim_{t\to\infty} \int_0^t e^{C(t-s)H^2} e^{C^T(t-s)} \mathrm{d}s = \int_0^\infty e^{Ct} H^2 e^{C^T t} \mathrm{d}t.$$

Additionally, we can get that the matrix Σ is at least positive semi-definite because H^2 is positive semi-definite. In the stationary case, that is, $t \to \infty$, then the transient distribution of the solution $X(t)$ will converge to the stationary normal distribution $\mathbb{N}(0, \Sigma)$, which gives that the stationary solution $(u_1(t), u_2(t), u_3(t), u_4(t), u_5(t))^T$ follows a unique normal probability density $\Phi(u_1, u_2, u_3, u_4, u_5)$ in a long term.

In addition, in light of the derivability of the matrix function $e^{Ct}H^2e^{C^Tt}$, one has

(i) $\displaystyle\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}(e^{Ct}H^2e^{C^Tt})\mathrm{d}t = C\Sigma + \Sigma C^T$;

(ii) $\displaystyle\int_0^\infty \frac{\mathrm{d}}{\mathrm{d}t}(e^{Ct}H^2e^{C^Tt})\mathrm{d}t = -H^2$, which gives that

$$H^2 + C\Sigma + \Sigma C^T = 0. \tag{4.4}$$

Now we are in the position to give the specific form of covariance matrix Σ. For this purpose, we define $C_1 = J_1 C J_1^{-1}$, where the ordering matrix J_1 is given by

$$J_1 = \begin{pmatrix} 0&0&0&0&1 \\ 0&0&1&0&0 \\ 1&0&0&0&0 \\ 0&1&0&0&0 \\ 0&0&0&1&0 \end{pmatrix}.$$

Then

$$C_1 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{33} & -a_{31} & a_{32} & 0 \\ -a_{15} & -a_{13} & -a_{11} & 0 & 0 \\ a_{25} & a_{23} & a_{21} & -a_{22} & -a_{24} \\ 0 & 0 & 0 & a_{42} & -a_{44} \end{pmatrix}.$$

Define $C_2 = J_2 C_1 J_2^{-1}$, where the elimination matrix J_2 takes the form

$$J_2 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

By simple computation, one can get that

$$C_2 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & -a_{31} & a_{32} & 0 \\ 0 & p_1 & a_{31} - a_{11} & -a_{32} & 0 \\ 0 & p_2 & 0 & a_{32} - a_{22} & -a_{24} \\ 0 & -a_{42} & 0 & a_{42} & -a_{44} \end{pmatrix},$$

where

$$\begin{aligned} p_1 &= a_{31} + a_{32} + a_{33} - a_{11} - a_{13} = m + \gamma - \alpha, \\ p_2 &= a_{22} + a_{23} - a_{32} - a_{33} = -\frac{\gamma V^*}{I^*} - \gamma \neq 0. \end{aligned}$$

Let $C_3 = J_3 C_2 J_3^{-1}$, where the ordering matrix J_3 is given by

$$J_3 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}.$$

Then

$$C_3 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & 0 & a_{32} & -a_{31} \\ 0 & -a_{42} & -a_{44} & a_{42} & 0 \\ 0 & p_2 & -a_{24} & a_{32} - a_{22} & 0 \\ 0 & p_1 & 0 & -a_{32} & a_{31} - a_{11} \end{pmatrix}.$$

According to the value of p_1, we will consider the following two cases: (1) $p_1 \neq 0$; (2) $p_1 = 0$.

Case 1. If $p_1 \neq 0$, let $C_4 = J_4 C_3 J_4^{-1}$, where the elimination matrix J_4 takes the form

$$J_4 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{p_2}{a_{42}} & 1 & 0 \\ 0 & 0 & \frac{p_1}{a_{42}} & 0 & 1 \end{pmatrix}.$$

Direct calculation leads to that

$$C_4 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & \frac{a_{31}p_1 - a_{32}p_2}{a_{42}} & a_{32} & -a_{31} \\ 0 & -a_{42} & -a_{44} - p_2 & a_{42} & 0 \\ 0 & 0 & q_1 & p_2 + a_{32} - a_{22} & 0 \\ 0 & 0 & q_2 & p_1 - a_{32} & a_{31} - a_{11} \end{pmatrix},$$

where

$$q_1 = -\frac{p_2(a_{44} + p_2 + a_{32} - a_{22}) + a_{24}a_{42}}{a_{42}},$$
$$q_2 = -\frac{p_1(a_{31} + a_{44} - a_{11}) + p_2(p_1 - a_{32})}{a_{42}}.$$

On the basis of the values of q_1 and q_2, we will investigate the following four subcases: (i) $q_1 \neq 0, q_2 \neq 0$; (ii) $q_1 = 0, q_2 \neq 0$; (iii) $q_1 \neq 0, q_2 = 0$; (iv) $q_1 = 0, q_2 = 0$.

Case 1.1. If $q_1 \neq 0, q_2 \neq 0$, let $C_5 = J_5 C_4 J_5^{-1}$, where the elimination matrix J_5 is given by

$$J_5 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\frac{q_2}{q_1} & 1 \end{pmatrix}.$$

Then

$$C_5 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & \frac{a_{31}p_1 - a_{32}p_2}{a_{42}} & \frac{a_{32}q_1 - a_{31}q_2}{q_1} & -a_{31} \\ 0 & -a_{42} & -a_{44} - p_2 & a_{42} & 0 \\ 0 & 0 & q_1 & p_2 + a_{32} - a_{22} & 0 \\ 0 & 0 & 0 & r_1 & a_{31} - a_{11} \end{pmatrix},$$

where

$$r_1 = \frac{q_2(a_{22} + a_{31} - a_{11} - a_{32} - p_2)}{q_1} + p_1 - a_{32}.$$

Based on the value of r_1, we will analyze the following two conditions:
(a) $r_1 \neq 0$; (b) $r_1 = 0$.

Case 1.1.1. If $r_1 \neq 0$, let $C_6 = J_6 C_5 J_6^{-1}$, where the standardized transformation matrix J_6 takes the form

$$J_6 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{42}r_1q_1 & r_1q_1(a_{31} + a_{32} - a_{11} - a_{22} - a_{44}) & j_{6(24)} & (a_{31} - a_{11})^3 \\ 0 & 0 & r_1q_1 & r_1(a_{23} + a_{31} - a_{11} - a_{33}) & (a_{31} - a_{11})^2 \\ 0 & 0 & 0 & r_1 & a_{31} - a_{11} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

here

$$j_{6(24)} = r_1[a_{42}q_1 + (a_{23} + a_{31} - a_{11} - a_{33})(a_{23} - a_{33}) + (a_{31} - a_{11})^2].$$

Then

$$C_6 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ a_{35}a_{42}r_1q_1 & -(a_{11} + a_{22} + a_{33} + a_{44}) & c_{6(23)} & c_{6(24)} & c_{6(25)} \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

where

$$\begin{aligned} c_{6(23)} &= -\tau_2 \\ &= -(a_{11}a_{22} + a_{11}a_{33} + a_{11}a_{44} + a_{22}a_{33} + a_{22}a_{44} + a_{33}a_{44} \\ &\quad + a_{24}a_{42} - a_{13}a_{31} - a_{23}a_{32}), \\ c_{6(24)} &= -\tau_3 \\ &= -(a_{11}a_{22}a_{33} + a_{11}a_{22}a_{44} + a_{11}a_{33}a_{44} + a_{11}a_{24}a_{42} \\ &\quad + a_{13}a_{21}a_{32} + a_{22}a_{33}a_{44} + a_{24}a_{33}a_{42} - a_{11}a_{23}a_{32} \\ &\quad - a_{13}a_{22}a_{31} - a_{13}a_{31}a_{44} - a_{23}a_{32}a_{44}), \\ c_{6(25)} &= -\tau_4 \\ &= -(a_{11}a_{22}a_{33}a_{44} + a_{11}a_{24}a_{33}a_{42} + a_{13}a_{21}a_{32}a_{44} \\ &\quad - a_{11}a_{23}a_{32}a_{44} - a_{13}a_{22}a_{31}a_{44} - a_{13}a_{24}a_{31}a_{42}). \end{aligned}$$

Next, define $C_7 = J_7 C_6 J_7^{-1}$, where the standardized transformation matrix J_7 is given by

$$J_7 = \begin{pmatrix} a_{35}a_{42}r_1q_1 & -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

We can compute that

$$C_7 = \begin{pmatrix} -(\tau_1+\rho) & -(\tau_1\rho+\tau_2) & -(\tau_2\rho+\tau_3) & -(\tau_3\rho+\tau_4) & -\tau_4\rho \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix},$$

where τ_1, τ_2, τ_3 and τ_4 are the same as above. In this case, Eq. (4.4) can be transformed into the following form

$$\begin{aligned}&(J_7J_6J_5J_4J_3J_2J_1)H^2(J_7J_6J_5J_4J_3J_2J_1)^T + C_7(J_7J_6J_5J_4J_3J_2J_1)\Sigma(J_7J_6J_5J_4J_3J_2J_1)^T \\ &\quad + (J_7J_6J_5J_4J_3J_2J_1)\Sigma(J_7J_6J_5J_4J_3J_2J_1)^T C_7^T = 0,\end{aligned}$$

that is,

$$H_0^2 + C_7\Sigma_0 + \Sigma_0 C_7^T = 0,$$

where $H_0 = \mathrm{diag}(1,0,0,0,0)$, $\Sigma_0 = \varrho_1^{-2}(J_7J_6J_5J_4J_3J_2J_1)\Sigma(J_7J_6J_5J_4J_3J_2J_1)^T$, $\varrho_1 = a_{35}a_{42}r_1q_1\sigma$. Note that the matrix C has all negative real-part eigenvalues, then C_7 is a Hurwitz matrix. In view of Lemma 4.1, one can obtain that the matrix Σ_0 is positive definite whose exact form is

$$\Sigma_0 = \begin{pmatrix} \vartheta_{11} & 0 & -\vartheta_{22} & 0 & \vartheta_{33} \\ 0 & \vartheta_{22} & 0 & -\vartheta_{33} & 0 \\ -\vartheta_{22} & 0 & \vartheta_{33} & 0 & -\vartheta_{44} \\ 0 & -\vartheta_{33} & 0 & \vartheta_{44} & 0 \\ \vartheta_{33} & 0 & -\vartheta_{44} & 0 & \vartheta_{55} \end{pmatrix},$$

where

$$\begin{aligned}\vartheta_{11} &= \frac{(\tau_1\rho+\tau_2)[(\tau_2\rho+\tau_3)(\tau_3\rho+\tau_4)-(\tau_1\rho+\tau_2)\tau_4\rho]-(\tau_3\rho+\tau_4)[(\tau_1+\rho)(\tau_3\rho+\tau_4)-\tau_4\rho]}{2\Delta}, \\ \vartheta_{22} &= \frac{(\tau_2\rho+\tau_3)(\tau_3\rho+\tau_4)-(\tau_1\rho+\tau_2)\tau_4\rho}{2\Delta}, \\ \vartheta_{33} &= \frac{(\tau_1+\rho)(\tau_3\rho+\tau_4)-\tau_4\rho}{2\Delta},\end{aligned}$$

$$\vartheta_{44} = \frac{(\tau_1+\rho)(\tau_1\rho+\tau_2)-(\tau_2\rho+\tau_3)}{2\Delta},$$
$$\vartheta_{55} = \frac{(\tau_2\rho+\tau_3)[(\tau_1+\rho)(\tau_1\rho+\tau_2)-(\tau_2\rho+\tau_3)]-(\tau_1+\rho)[(\tau_1+\rho)(\tau_3\rho+\tau_4)-\tau_4\rho]}{2\tau_4\rho\Delta},$$

and

$$\Delta = [(\tau_1+\rho)(\tau_1\rho+\tau_2)-(\tau_2\rho+\tau_3)][(\tau_2\rho+\tau_3)(\tau_3\rho+\tau_4)-(\tau_1\rho+\tau_2)\tau_4\rho] - [(\tau_1+\rho)(\tau_3\rho+\tau_4)-\tau_4\rho]^2.$$

Hence, the matrix $\Sigma = \varrho_1^2(J_7J_6J_5J_4J_3J_2J_1)^{-1}\Sigma_0[(J_7J_6J_5J_4J_3J_2J_1)^{-1}]^T$ is also positive definite.

Case 1.1.2. If $r_1 = 0$, let $C_8 = J_8C_5J_8^{-1}$, where the standardized transformation matrix J_8 takes the form

$$J_8 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{42}q_1 & q_1(a_{32}-a_{22}-a_{44}) & a_{42}q_1+(p_2+a_{32}-a_{22})^2 & 0 \\ 0 & 0 & q_1 & p_2+a_{32}-a_{22} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_8 = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ a_{35}a_{42}q_1 & -l_1 & -l_2 & -l_3 & a_{31}a_{42}q_1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{31}-a_{11} \end{pmatrix},$$

where

$$l_1 = a_{22}+a_{31}+a_{33}+a_{44},\ l_2 = a_{42}q_1 - a_{31}p_1 + a_{32}p_2 + (a_{32}-a_{22}-a_{44})(a_{22}+a_{31}+a_{33}-p_2) + (p_2+a_{32}-a_{22})^2,$$
$$l_3 = a_{42}q_1(a_{31}+a_{33}) + a_{31}a_{42}q_2 + (p_2+a_{32}-a_{22}) [(a_{44}+p_2)(a_{31}+a_{32}+a_{33}) + a_{31}p_1 - a_{32}p_2].$$

Next, define $C_9 = J_9C_8J_9^{-1}$, where the standardized transformation matrix J_9 is given by

$$J_9 = \begin{pmatrix} a_{35}a_{42}q_1 & -l_1 & -l_2 & -l_3 & a_{31}a_{42}q_1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

By simple calculation, one has

$$C_9 = \begin{pmatrix} -(\rho + l_1) & -(\rho l_1 + l_2) & -(\rho l_2 + l_3) & -\rho l_3 & a_{31}a_{42}q_1(a_{31} - a_{11} + \rho) \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{31} - a_{11} \end{pmatrix}.$$

In this case, Eq. (4.4) can be rewritten in the following form

$$\begin{aligned} &(J_9J_8J_5J_4J_3J_2J_1)H^2(J_9J_8J_5J_4J_3J_2J_1)^T + C_9(J_9J_8J_5J_4J_3J_2J_1)\Sigma(J_9J_8J_5J_4J_3J_2J_1)^T \\ &+ (J_9J_8J_5J_4J_3J_2J_1)\Sigma(J_9J_8J_5J_4J_3J_2J_1)^T C_9^T = 0, \end{aligned}$$

i.e.,

$$H_0^2 + C_9\Theta_0 + \Theta_0 C_9^T = 0, \tag{4.5}$$

where $\Theta_0 = \varrho_2^{-2}(J_9J_8J_5J_4J_3J_2J_1)\Sigma(J_9J_8J_5J_4J_3J_2J_1)^T$, $\varrho_2 = a_{35}a_{42}q_1\sigma$. By solving Eq. (4.5), we get

$$\Theta_0 = \begin{pmatrix} \frac{\rho^2(l_1l_2 - l_3) + l_2(\rho l_2 + l_3)}{\Sigma_{01}^*} & 0 & -\frac{\rho l_2 + l_3}{\Sigma_{01}^*} & 0 & 0 \\ 0 & \frac{\rho l_2 + l_3}{\Sigma_{01}^*} & 0 & -\frac{\rho + l_1}{\Sigma_{01}^*} & 0 \\ -\frac{\rho l_2 + l_3}{\Sigma_{01}^*} & 0 & \frac{\rho + l_1}{\Sigma_{01}^*} & 0 & 0 \\ 0 & -\frac{\rho + l_1}{\Sigma_{01}^*} & 0 & \frac{\rho l_1(\rho + l_1) + l_1l_2 - l_3}{\rho l_3 \Sigma_{01}^*} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where

$$\Sigma_{01}^* = 2(\rho^3 + \rho^2 l_1 + \rho l_2 + l_3)(l_1l_2 - l_3).$$

As we can see that the matrix Θ_0 is positive semi-definite and its submatrix

$$\Theta_0^{(4)} = \begin{pmatrix} \frac{\rho^2(l_1l_2 - l_3) + l_2(\rho l_2 + l_3)}{\Sigma_{01}^*} & 0 & -\frac{\rho l_2 + l_3}{\Sigma_{01}^*} & 0 \\ 0 & \frac{\rho l_2 + l_3}{\Sigma_{01}^*} & 0 & -\frac{\rho + l_1}{\Sigma_{01}^*} \\ -\frac{\rho l_2 + l_3}{\Sigma_{01}^*} & 0 & \frac{\rho + l_1}{\Sigma_{01}^*} & 0 \\ 0 & -\frac{\rho + l_1}{\Sigma_{01}^*} & 0 & \frac{\rho l_1(\rho + l_1) + l_1l_2 - l_3}{\rho l_3 \Sigma_{01}^*} \end{pmatrix}$$

is positive definite. As a result, the matrix $\Sigma = \varrho_2^2(J_9J_8J_5J_4J_3J_2J_1)^{-1}\Theta_0[(J_9J_8J_5J_4J_3J_2J_1)^{-1}]^T$ is also positive semi-definite.

Case 1.2. If $q_1 = 0$, $q_2 \neq 0$, let $C_{10} = J_{10}C_4J_{10}^{-1}$, where the ordering matrix J_{10} takes the form

$$J_{10} = \begin{pmatrix} 1&0&0&0&0\\ 0&1&0&0&0\\ 0&0&1&0&0\\ 0&0&0&0&1\\ 0&0&0&1&0 \end{pmatrix}.$$

Then

$$C_{10} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0\\ -a_{35} & -a_{31}-a_{32}-a_{33} & \dfrac{a_{31}p_1 - a_{32}p_2}{a_{42}} & -a_{31} & a_{32}\\ 0 & -a_{42} & -a_{44}-p_2 & 0 & a_{42}\\ 0 & 0 & q_2 & a_{31}-a_{11} & p_1-a_{32}\\ 0 & 0 & 0 & 0 & p_2+a_{32}-a_{22} \end{pmatrix}.$$

Next, define $C_{11} = J_{11}C_{10}J_{11}^{-1}$, where the standardized transformation matrix J_{11} is given by

$$J_{11} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0\\ 0 & -a_{42}q_2 & q_2(a_{31}-a_{11}-a_{44}-p_2) & (a_{31}-a_{11})^2 & a_{42}q_2+(p_1-a_{32})(a_{23}+a_{31}-a_{11}-a_{33})\\ 0 & 0 & q_2 & a_{31}-a_{11} & p_1-a_{32}\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{11} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0\\ a_{35}a_{42}q_2 & -\tilde{l}_1 & -\tilde{l}_2 & -\tilde{l}_3 & -\tilde{l}_4\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & p_2+a_{32}-a_{22} \end{pmatrix},$$

where

$$\begin{aligned}
\tilde{l}_1 &= a_{11}+a_{32}+a_{33}+a_{44}+p_2,\\
\tilde{l}_2 &= a_{31}p_1+(a_{11}+a_{32}+a_{33})(a_{11}+a_{44}+p_2-a_{31})-a_{32}p_2-(a_{31}-a_{11})^2,\\
\tilde{l}_3 &= (a_{11}-a_{31})(a_{31}p_1-a_{32}p_2)-a_{31}a_{42}q_2-(a_{31}-a_{11})(a_{44}+p_2)(a_{31}+a_{32}+a_{33}),\\
\tilde{l}_4 &= a_{42}q_2(a_{22}-a_{31}-a_{32}-a_{33}-p_2)+(a_{32}-p_1)(a_{44}+p_2)(a_{31}+a_{32}+a_{33})\\
&\quad +(a_{32}-p_1)(a_{31}p_1-a_{32}p_2)+(p_1-a_{32})(a_{33}-a_{23})(a_{11}+a_{32}+a_{33}+a_{44}+p_2)\\
&\quad +(a_{32}-p_1)(p_2+a_{32}-a_{22})(a_{23}+a_{31}-a_{11}-a_{33}).
\end{aligned}$$

Define $C_{12} = J_{12}C_{11}J_{12}^{-1}$, where the standardized transformation matrix J_{12} takes the form

$$J_{12} = \begin{pmatrix} a_{35}a_{42}q_2 & -\tilde{l}_1 & -\tilde{l}_2 & -\tilde{l}_3 & -\tilde{l}_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{12} = \begin{pmatrix} -(\rho+\tilde{l}_1) & -(\rho\tilde{l}_1+\tilde{l}_2) & -(\rho\tilde{l}_2+\tilde{l}_3) & -\rho\tilde{l}_3 & -\tilde{l}_4(p_2+a_{32}-a_{22}+\rho) \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & p_2+a_{32}-a_{22} \end{pmatrix}.$$

In this case, Eq. (4.4) becomes the following form

$$\begin{aligned} &(J_{12}J_{11}J_{10}J_4J_3J_2J_1)H^2(J_{12}J_{11}J_{10}J_4J_3J_2J_1)^T \\ &\quad + C_{12}(J_{12}J_{11}J_{10}J_4J_3J_2J_1)\Sigma(J_{12}J_{11}J_{10}J_4J_3J_2J_1)^T \\ &\quad + (J_{12}J_{11}J_{10}J_4J_3J_2J_1)\Sigma(J_{12}J_{11}J_{10}J_4J_3J_2J_1)^T C_{12}^T = 0, \end{aligned}$$

that is,

$$H_0^2 + C_{12}\tilde{\Theta}_0 + \tilde{\Theta}_0 C_{12}^T = 0, \tag{4.6}$$

where $\tilde{\Theta}_0 = \varrho_3^{-2}(J_{12}J_{11}J_{10}J_4J_3J_2J_1)\Sigma(J_{12}J_{11}J_{10}J_4J_3J_2J_1)^T$, $\varrho_3 = a_{35}a_{42}q_2\sigma$. By solving Eq. (4.6), we have

$$\tilde{\Theta}_0 = \begin{pmatrix} \frac{\rho^2(\tilde{l}_1\tilde{l}_2-\tilde{l}_3)+\tilde{l}_2(\rho\tilde{l}_2+\tilde{l}_3)}{\Sigma_{02}^*} & 0 & -\frac{\rho\tilde{l}_2+\tilde{l}_3}{\Sigma_{02}^*} & 0 & 0 \\ 0 & \frac{\rho\tilde{l}_2+\tilde{l}_3}{\Sigma_{02}^*} & 0 & -\frac{\rho+\tilde{l}_1}{\Sigma_{02}^*} & 0 \\ -\frac{\rho\tilde{l}_2+\tilde{l}_3}{\Sigma_{02}^*} & 0 & \frac{\rho+\tilde{l}_1}{\Sigma_{02}^*} & 0 & 0 \\ 0 & -\frac{\rho+\tilde{l}_1}{\Sigma_{02}^*} & 0 & \frac{\rho\tilde{l}_1(\rho+\tilde{l}_1)+\tilde{l}_1\tilde{l}_2-\tilde{l}_3}{\rho\tilde{l}_3\Sigma_{02}^*} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where

$$\Sigma_{02}^* = 2(\rho^3+\rho^2\tilde{l}_1+\rho\tilde{l}_2+\tilde{l}_3)(\tilde{l}_1\tilde{l}_2-\tilde{l}_3).$$

As we can see that the matrix $\tilde{\Theta}_0$ is positive semi-definite and its submatrix

$$\tilde{\Theta}_0^{(4)} = \begin{pmatrix} \frac{\rho^2(\tilde{l}_1\tilde{l}_2 - \tilde{l}_3) + \tilde{l}_2(\rho\tilde{l}_2 + \tilde{l}_3)}{\Sigma_{02}^*} & 0 & -\frac{\rho\tilde{l}_2 + \tilde{l}_3}{\Sigma_{02}^*} & 0 \\ 0 & \frac{\rho\tilde{l}_2 + \tilde{l}_3}{\Sigma_{02}^*} & 0 & -\frac{\rho + \tilde{l}_1}{\Sigma_{02}^*} \\ -\frac{\rho\tilde{l}_2 + \tilde{l}_3}{\Sigma_{02}^*} & 0 & \frac{\rho + \tilde{l}_1}{\Sigma_{02}^*} & 0 \\ 0 & -\frac{\rho + \tilde{l}_1}{\Sigma_{02}^*} & 0 & \frac{\rho\tilde{l}_1(\rho + \tilde{l}_1) + \tilde{l}_1\tilde{l}_2 - \tilde{l}_3}{\rho\tilde{l}_3\Sigma_{02}^*} \end{pmatrix}$$

is positive definite. Consequently, the matrix $\Sigma = \varrho_3^2(J_{12}J_{11}J_{10}J_4J_3J_2J_1)^{-1}\tilde{\Theta}_0[(J_{12}J_{11}J_{10}J_4J_3J_2J_1)^{-1}]^T$ is also positive semi-definite.

Case 1.3. If $q_1 \neq 0$, $q_2 = 0$ and $p_1 \neq a_{32}$, let $C_{13} = J_{13}C_4J_{13}^{-1}$, where the standardized transformation matrix J_{13} is given by

$$J_{13} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{42}q_1(p_1 - a_{32}) & j_{13(23)} & j_{13(24)} & (a_{31} - a_{11})^3 \\ 0 & 0 & q_1(p_1 - a_{32}) & (p_1 - a_{32})(a_{23} + a_{31} - a_{11} - a_{33}) & (a_{31} - a_{11})^2 \\ 0 & 0 & 0 & p_1 - a_{32} & a_{31} - a_{11} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

here

$$\begin{aligned} j_{13(23)} &= q_1(p_1 - a_{32})(a_{31} + a_{32} - a_{11} - a_{22} - a_{44}), \\ j_{13(24)} &= (p_1 - a_{32})[a_{42}q_1 + (a_{23} \\ &\quad + a_{31} - a_{11} - a_{33})(p_2 + a_{32} - a_{22}) + (a_{31} - a_{11})^2]. \end{aligned}$$

Then

$$C_{13} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ a_{35}a_{42}q_1(p_1 - a_{32}) & -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Next, let $C_{14} = J_{14}C_{13}J_{14}^{-1}$, where the standardized transformation matrix J_{14} takes the form

$$J_{14} = \begin{pmatrix} a_{35}a_{42}q_1(p_1 - a_{32}) & -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{14} = C_7 = \begin{pmatrix} -(\tau_1+\rho) & -(\tau_1\rho+\tau_2) & -(\tau_2\rho+\tau_3) & -(\tau_3\rho+\tau_4) & -\tau_4\rho \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

In this case, Eq. (4.4) can be transformed into the following form

$$\begin{aligned}&(J_{14}J_{13}J_4J_3J_2J_1)H^2(J_{14}J_{13}J_4J_3J_2J_1)^T + C_{14}(J_{14}J_{13}J_4J_3J_2J_1)\Sigma(J_{14}J_{13}J_4J_3J_2J_1)^T \\ &\quad + (J_{14}J_{13}J_4J_3J_2J_1)\Sigma(J_{14}J_{13}J_4J_3J_2J_1)^T C_{14}^T = 0,\end{aligned}$$

i.e.,

$$H_0^2 + C_{14}\Sigma_0 + \Sigma_0 C_{14}^T = 0,$$

where $\Sigma_0 = \varrho_4^{-2}(J_{14}J_{13}J_4J_3J_2J_1)\Sigma(J_{14}J_{13}J_4J_3J_2J_1)^T$, $\varrho_4 = a_{35}a_{42}q_1(p_1-a_{32})\sigma$. So the matrix $\Sigma = \varrho_4^2(J_{14}J_{13}J_4J_3J_2J_1)^{-1}\Sigma_0[(J_{14}J_{13}J_4J_3J_2J_1)^{-1}]^T$ is also positive definite.

Case 1.3'. If $q_1 \neq 0$, $q_2 = 0$ and $p_1 = a_{32}$, let $C_{15} = J_{15}C_4J_{15}^{-1}$, where the standardized transformation matrix J_{15} is given by

$$J_{15} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{42}q_1 & q_1(a_{32}-a_{22}-a_{44}) & a_{42}q_1 + (p_2+a_{32}-a_{22})^2 & 0 \\ 0 & 0 & q_1 & p_2+a_{32}-a_{22} & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{15} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ a_{35}a_{42}q_1 & -\bar{l}_1 & -\bar{l}_2 & -\bar{l}_3 & a_{31}a_{42}q_1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{31}-a_{11} \end{pmatrix},$$

where

$$\begin{aligned}\bar{l}_1 &= a_{22}+a_{31}+a_{33}+a_{44}, \\ \bar{l}_2 &= a_{31}p_1 + (a_{22}+a_{44}-a_{32})(a_{22}+a_{31}+a_{33}-p_2) - a_{32}p_2 - a_{42}q_1 - (p_2+a_{32}-a_{22})^2, \\ \bar{l}_3 &= (a_{22}-a_{32}-p_2)[(p_2+a_{44})(a_{31}+a_{32}+a_{33}) + a_{31}p_1 - a_{32}p_2] - a_{42}q_1(a_{31}+a_{33}).\end{aligned}$$

Define $C_{16} = J_{16}C_{15}J_{16}^{-1}$, where the standardized transformation matrix J_{16} takes the form

$$J_{16} = \begin{pmatrix} a_{35}a_{42}q_1 & -\bar{l}_1 & -\bar{l}_2 & -\bar{l}_3 & a_{31}a_{42}q_1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{16} = \begin{pmatrix} -(\rho+\bar{l}_1) & -(\rho\bar{l}_1+\bar{l}_2) & -(\rho\bar{l}_2+\bar{l}_3) & -\rho\bar{l}_3 & a_{31}a_{42}q_1(a_{31}-a_{11}+\rho) \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{31}-a_{11} \end{pmatrix}.$$

In this case, Eq. (4.4) becomes the following form

$$\begin{aligned}&(J_{16}J_{15}J_4J_3J_2J_1)H^2(J_{16}J_{15}J_4J_3J_2J_1)^T + C_{16}(J_{16}J_{15}J_4J_3J_2J_1)\Sigma(J_{16}J_{15}J_4J_3J_2J_1)^T \\ &\quad + (J_{16}J_{15}J_4J_3J_2J_1)\Sigma(J_{16}J_{15}J_4J_3J_2J_1)^T C_{16}^T = 0,\end{aligned}$$

that is,

$$H_0^2 + C_{16}\bar{\Theta}_0 + \bar{\Theta}_0 C_{16}^T = 0, \tag{4.7}$$

where $\bar{\Theta}_0 = \varrho_5^{-2}(J_{16}J_{15}J_4J_3J_2J_1)\Sigma(J_{16}J_{15}J_4J_3J_2J_1)^T$, $\varrho_5 = a_{35}a_{42}q_1\sigma$. By solving Eq. (4.7), we get

$$\bar{\Theta}_0 = \begin{pmatrix} \frac{\rho^2(\bar{l}_1\bar{l}_2-\bar{l}_3)+\bar{l}_2(\rho\bar{l}_2+\bar{l}_3)}{\Sigma_{03}^*} & 0 & -\frac{\rho\bar{l}_2+\bar{l}_3}{\Sigma_{03}^*} & 0 & 0 \\ 0 & \frac{\rho\bar{l}_2+\bar{l}_3}{\Sigma_{03}^*} & 0 & -\frac{\rho+\bar{l}_1}{\Sigma_{03}^*} & 0 \\ -\frac{\rho\bar{l}_2+\bar{l}_3}{\Sigma_{03}^*} & 0 & \frac{\rho+\bar{l}_1}{\Sigma_{03}^*} & 0 & 0 \\ 0 & -\frac{\rho+\bar{l}_1}{\Sigma_{03}^*} & 0 & \frac{\rho\bar{l}_1(\rho+\bar{l}_1)+\bar{l}_1\bar{l}_2-\bar{l}_3}{\rho\bar{l}_3\Sigma_{03}^*} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where

$$\Sigma_{03}^* = 2(\rho^3+\rho^2\bar{l}_1+\rho\bar{l}_2+\bar{l}_3)(\bar{l}_1\bar{l}_2-\bar{l}_3).$$

Obviously, the matrix $\bar{\Theta}_0$ is positive semi-definite and its submatrix

$$\bar{\Theta}_0^{(4)} = \begin{pmatrix} \frac{\rho^2(\bar{l}_1\bar{l}_2 - \bar{l}_3) + \bar{l}_2(\rho\bar{l}_2 + \bar{l}_3)}{\Sigma_{03}^*} & 0 & -\frac{\rho\bar{l}_2 + \bar{l}_3}{\Sigma_{03}^*} & 0 \\ 0 & \frac{\rho\bar{l}_2 + \bar{l}_3}{\Sigma_{03}^*} & 0 & -\frac{\rho + \bar{l}_1}{\Sigma_{03}^*} \\ -\frac{\rho\bar{l}_2 + \bar{l}_3}{\Sigma_{03}^*} & 0 & \frac{\rho + \bar{l}_1}{\Sigma_{03}^*} & 0 \\ 0 & -\frac{\rho + \bar{l}_1}{\Sigma_{03}^*} & 0 & \frac{\rho\bar{l}_1(\rho + \bar{l}_1) + \bar{l}_1\bar{l}_2 - \bar{l}_3}{\rho\bar{l}_3\Sigma_{03}^*} \end{pmatrix}$$

is positive definite. As a result, the matrix $\Sigma = \varrho_5^2(J_{16}J_{15}J_4J_3J_2J_1)^{-1}\bar{\Theta}_0[(J_{16}J_{15}J_4J_3J_2J_1)^{-1}]^T$ is also positive semi-definite.

Case 1.4. If $q_1 = 0, q_2 = 0$, then Eq. (4.4) can be rewritten in the form:

$$\begin{aligned} &(J_4J_3J_2J_1)H^2(J_4J_3J_2J_1)^T + C_4(J_4J_3J_2J_1)\Sigma(J_4J_3J_2J_1)^T \\ &\quad + (J_4J_3J_2J_1)\Sigma(J_4J_3J_2J_1)^T C_4^T = 0. \end{aligned} \tag{4.8}$$

Define

$$F_1 = \begin{pmatrix} -\rho & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & \frac{a_{31}p_1 - a_{32}p_2}{a_{42}} \\ 0 & -a_{42} & -(a_{44} + p_2) \end{pmatrix}, F_2 = \begin{pmatrix} p_2 + a_{32} - a_{22} & 0 \\ p_1 - a_{32} & a_{31} - a_{11} \end{pmatrix},$$

$$F_3 = \begin{pmatrix} 0 & 0 \\ a_{32} & -a_{31} \\ a_{42} & 0 \end{pmatrix},$$

then the characteristic polynomial of C is

$$\begin{aligned} \varphi_C(\lambda) &= |\lambda\mathbf{I}_3 - F_1||\lambda\mathbf{I}_2 - F_2| \\ &:= (\lambda^3 + \zeta_1\lambda^2 + \zeta_2\lambda + \zeta_3)[\lambda^2 + (a_{11} + a_{33} - a_{23} - a_{31})\lambda + (a_{31} - a_{11})(p_2 + a_{32} - a_{22})], \end{aligned}$$

where

$$\begin{aligned} \zeta_1 &= \rho + a_{22} + a_{23} + a_{31} + a_{44}, \\ \zeta_2 &= \rho(a_{22} + a_{23} + a_{31} + a_{44}) + a_{44}(a_{31} + a_{32} + a_{33}) \\ &\quad + a_{31}p_1 + a_{31}p_2 + a_{33}p_2, \\ \zeta_3 &= \rho(a_{31}a_{44} + a_{32}a_{44} + a_{33}a_{44} + a_{31}p_1 + a_{31}p_2 + a_{33}p_2). \end{aligned}$$

Since C is a Hurwitz matrix, then all of the roots of equation $\lambda^3 + \zeta_1\lambda^2 + \zeta_2\lambda + \zeta_3 = 0$ have negative real part. In light of Definition 4.1, one has

$$\zeta_1 > 0, \ \zeta_3 > 0 \text{ and } \zeta_1\zeta_2 - \zeta_3 > 0. \tag{4.9}$$

Define two nonsingular transformed matrices P_1 and Q_1 as follows

$$P_1 = \begin{pmatrix} a_{35}a_{42} & a_{42}(a_{22}+a_{23}+a_{31}+a_{44}) & (a_{44}+p_2)^2 + a_{32}p_2 - a_{31}p_1 \\ 0 & -a_{42} & -(a_{44}+p_2) \\ 0 & 0 & 1 \end{pmatrix},$$
$$Q_1 = \begin{pmatrix} P_1 & \mathbb{O}_{3,2} \\ \mathbb{O}_{2,3} & \mathbf{I}_2 \end{pmatrix}.$$

Let the matrices $F_4 = P_1 F_3 := (\tilde{c}_{ij})_{1\le i\le 3, 4\le j\le 5}$ and $R_1 = Q_1 C_4 Q_1^{-1}$. Since

$$Q_1^{-1} = \begin{pmatrix} P_1^{-1} & \mathbb{O}_{3,2} \\ \mathbb{O}_{2,3} & \mathbf{I}_2 \end{pmatrix},$$

we get $(Q_1 J_4 J_3 J_2 J_1) H^2 (Q_1 J_4 J_3 J_2 J_1)^T = \mathrm{diag}(a_{35}^2 a_{42}^2 \sigma^2, 0, 0, 0, 0) = (a_{35}a_{42}\sigma)^2 H^2$ and

$$R_1 = \begin{pmatrix} P_1 F_1 P_1^{-1} & F_4 \\ \mathbb{O}_{2,3} & F_2 \end{pmatrix} = \begin{pmatrix} -\zeta_1 & -\zeta_2 & -\zeta_3 & 0 & 0 \\ 1 & 0 & 0 & \tilde{a}_{32} & -\tilde{a}_{31} \\ 0 & 1 & 0 & \tilde{a}_{42} & 0 \\ 0 & 0 & 0 & p_2 + a_{32} - a_{22} & 0 \\ 0 & 0 & 0 & p_1 - a_{32} & a_{31} - a_{11} \end{pmatrix}.$$

Accordingly, Eq. (4.8) can be rewritten in the form

$$\begin{aligned} &(J_4 J_3 J_2 J_1) H^2 (J_4 J_3 J_2 J_1)^T + R_1 \left[\frac{1}{(a_{35}a_{42}\sigma)^2} (Q_1 J_4 J_3 J_2 J_1) \Sigma (Q_1 J_4 J_3 J_2 J_1)^T \right] \\ &+ \left[\frac{1}{(a_{35}a_{42}\sigma)^2} (Q_1 J_4 J_3 J_2 J_1) \Sigma (Q_1 J_4 J_3 J_2 J_1)^T \right] R_1^T = 0. \end{aligned} \tag{4.10}$$

The solution of Eq. (4.10) is unique which satisfies

$$\begin{aligned} &\frac{1}{(a_{35}a_{42}\sigma)^2} (Q_1 J_4 J_3 J_2 J_1) \Sigma (Q_1 J_4 J_3 J_2 J_1)^T \\ &= \begin{pmatrix} \frac{\zeta_2}{2(\zeta_1\zeta_2 - \zeta_3)} & 0 & -\frac{1}{2(\zeta_1\zeta_2 - \zeta_3)} & 0 & 0 \\ 0 & \frac{1}{2(\zeta_1\zeta_2 - \zeta_3)} & 0 & 0 & 0 \\ -\frac{1}{2(\zeta_1\zeta_2 - \zeta_3)} & 0 & \frac{\zeta_1}{2\zeta_3(\zeta_1\zeta_2 - \zeta_3)} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} := \tilde{\Sigma}_0. \end{aligned}$$

On the basis of (4.9) and Han et al. (2020), one can see that $\tilde{\Sigma}_0$ is a positive semi-definite matrix. Thereby, the matrix $\Sigma = (a_{35}a_{42}\sigma)^2 (Q_1 J_4 J_3 J_2 J_1)^{-1} \tilde{\Sigma}_0 [(Q_1 J_4 J_3 J_2 J_1)^{-1}]^T$ is also positive semi-definite.

Case 2. If $p_1 = 0$, let $C_{17} = J_{17} C_3 J_{17}^{-1}$, where the elimination matrix J_{17} is given by

$$J_{17} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{p_2}{a_{42}} & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

With the aid of direct calculation, we obtain

$$C_{17} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & -\frac{a_{32} p_2}{a_{42}} & a_{32} & -a_{31} \\ 0 & -a_{42} & -a_{44} - p_2 & a_{42} & 0 \\ 0 & 0 & \lambda_1 & p_2 + a_{32} - a_{22} & 0 \\ 0 & 0 & \frac{a_{32} p_2}{a_{42}} & -a_{32} & a_{31} - a_{11} \end{pmatrix},$$

where

$$\lambda_1 = -\frac{p_2}{a_{42}}(a_{32} + a_{44} + p_2 - a_{22}) - a_{24}.$$

Next, define $C_{18} = J_{18} C_{17} J_{18}^{-1}$, where the ordering matrix J_{18} takes the form

$$J_{18} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Then

$$C_{18} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31} - a_{32} - a_{33} & -\frac{a_{32} p_2}{a_{42}} & -a_{31} & a_{32} \\ 0 & -a_{42} & -a_{44} - p_2 & 0 & a_{42} \\ 0 & 0 & \frac{a_{32} p_2}{a_{42}} & a_{31} - a_{11} & -a_{32} \\ 0 & 0 & \lambda_1 & 0 & p_2 + a_{32} - a_{22} \end{pmatrix}.$$

On the basis of the value of λ_1, we will study the following two conditions:
(i) $\lambda_1 \neq 0$; (ii) $\lambda_1 = 0$.

Case 2.1. If $\lambda_1 \neq 0$, let $C_{19} = J_{19}C_{18}J_{19}^{-1}$, where the elimination matrix J_{19} is given by

$$J_{19} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -\dfrac{a_{42}\lambda_1}{a_{32}p_2} & 1 \end{pmatrix}.$$

Then

$$C_{19} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ -a_{35} & -a_{31}-a_{32}-a_{33} & -\dfrac{a_{32}p_2}{a_{42}} & \dfrac{a_{42}\lambda_1}{p_2} - a_{31} & a_{32} \\ 0 & -a_{42} & -a_{44}-p_2 & \dfrac{a_{42}^2\lambda_1}{a_{32}p_2} & a_{42} \\ 0 & 0 & \dfrac{a_{32}p_2}{a_{42}} & a_{31}-a_{11}-\dfrac{a_{42}\lambda_1}{p_2} & -a_{32} \\ 0 & 0 & 0 & \xi_1 & p_2+a_{32}-a_{22}+\dfrac{a_{42}\lambda_1}{p_2} \end{pmatrix},$$

where

$$\xi_1 = \frac{a_{42}\lambda_1}{a_{32}p_2}\left(\frac{a_{42}\lambda_1}{p_2} + a_{11} + a_{23} - a_{31} - a_{33}\right).$$

According to the value of ξ_1, we will consider the following two conditions:

(a) $\xi_1 \neq 0$; (b) $\xi_1 = 0$.

Case 2.1.1. If $\xi_1 \neq 0$, let $C_{20} = J_{20}C_{19}J_{20}^{-1}$, where the standard transformation matrix J_{20} takes the form

$$J_{20} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{32}p_2\xi_1 & j_{20(23)} & j_{20(24)} & j_{20(25)} \\ 0 & 0 & \dfrac{a_{32}p_2\xi_1}{a_{42}} & \xi_1(a_{23}+a_{31}-a_{11}-a_{33}) & \left(p_2+a_{32}-a_{22}+\dfrac{a_{42}\lambda_1}{p_2}\right)^2 - a_{32}\xi_1 \\ 0 & 0 & 0 & \xi_1 & p_2+a_{32}-a_{22}+\dfrac{a_{42}\lambda_1}{p_2} \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

here

$$j_{20(23)} = \frac{a_{32}p_2\xi_1}{a_{42}}(a_{31}+a_{32}-a_{11}-a_{22}-a_{44}),$$

$$j_{20(24)} = \xi_1\left[a_{42}\lambda_1 + (a_{23}+a_{31}-a_{11}-a_{33})\left(a_{31}-a_{11}-\frac{a_{42}\lambda_1}{p_2}\right)\right.$$

$$+\left(p_2+a_{32}-a_{22}+\frac{a_{42}\lambda_1}{p_2}\right)^2-a_{32}\xi_1\Bigg],$$

$$j_{20(25)}=a_{32}\xi_1(a_{11}+a_{22}-a_{31}-a_{32})+\left(p_2+a_{32}-a_{22}+\frac{a_{42}\lambda_1}{p_2}\right)$$
$$\left[\left(p_2+a_{32}-a_{22}+\frac{a_{42}\lambda_1}{p_2}\right)^2-a_{32}\xi_1\right].$$

Then

$$C_{20}=\begin{pmatrix}-\rho & 0 & 0 & 0 & 0\\ a_{32}a_{35}p_2\xi_1 & -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\end{pmatrix}.$$

Next, define $C_{21}=J_{21}C_{20}J_{21}^{-1}$, where the standard transformation matrix J_{21} is given by

$$J_{21}=\begin{pmatrix}a_{32}a_{35}p_2\xi_1 & -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 1\end{pmatrix}.$$

Then

$$C_{21}=C_7=\begin{pmatrix}-(\tau_1+\rho) & -(\tau_1\rho+\tau_2) & -(\tau_2\rho+\tau_3) & -(\tau_3\rho+\tau_4) & -\tau_4\rho\\ 1 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1 & 0\end{pmatrix}.$$

In this case, Eq. (4.4) can be rewritten in the form

$$\begin{aligned}&(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)H^2(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)^T\\ &\quad+C_{21}(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)^T\\ &\quad+(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)^TC_{21}^T=0,\end{aligned}$$

i.e.,

$$H_0^2+C_{21}\Sigma_0+\Sigma_0C_{21}^T=0,$$

where $\Sigma_0=\varrho_6^{-2}(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)^T$, $\varrho_6=a_{32}a_{35}p_2\xi_1\sigma$. Consequently, the matrix $\Sigma=$

$\varrho_6^2(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)^{-1}\Sigma_0[(J_{21}J_{20}J_{19}J_{18}J_{17}J_3J_2J_1)^{-1}]^T$ is also positive definite.

Case 2.1.2. If $\xi_1 = 0$, let $C_{22} = J_{22}C_{19}J_{22}^{-1}$, where the standard transformation matrix J_{22} takes the form

$$J_{22} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{32}p_2 & \frac{a_{32}p_2}{a_{42}}\left(a_{31} - a_{11} - a_{44} - p_2 - \frac{a_{42}\lambda_1}{p_2}\right) & a_{42}\lambda_1 + \left(a_{31} - a_{11} - \frac{a_{42}\lambda_1}{p_2}\right)^2 & j_{22(25)} \\ 0 & 0 & \frac{a_{32}p_2}{a_{42}} & a_{31} - a_{11} - \frac{a_{42}\lambda_1}{p_2} & -a_{32} \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

here

$$j_{22(25)} = a_{32}(a_{11} + a_{22} - a_{31} - a_{32}).$$

Then

$$C_{22} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ a_{32}a_{35}p_2 & -\check{l}_1 & -\check{l}_2 & -\check{l}_3 & -\check{l}_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & p_2 + a_{32} - a_{22} + \frac{a_{42}\lambda_1}{p_2} \end{pmatrix},$$

where

$$\begin{aligned}
\check{l}_1 &= a_{11} + a_{33} + a_{44} + a_{32} + p_2 + \frac{a_{42}\lambda_1}{p_2}, \\
\check{l}_2 &= \left(a_{11} + a_{32} + a_{33} + \frac{a_{42}\lambda_1}{p_2}\right)\left(a_{11} + a_{44} + p_2 + \frac{a_{42}\lambda_1}{p_2} - a_{31}\right) \\
&\quad -a_{32}p_2 - a_{42}\lambda_1 - \left(a_{31} - a_{11} - \frac{a_{42}\lambda_1}{p_2}\right)^2, \\
\check{l}_3 &= a_{42}\lambda_1\left(a_{11} + \frac{a_{42}\lambda_1}{p_2} - 2a_{31} - a_{33}\right) + \left(a_{11} + \frac{a_{42}\lambda_1}{p_2} - a_{31}\right) \\
&\quad [(a_{44} + p_2)(a_{31} + a_{32} + a_{33}) - a_{32}p_2 - a_{42}\lambda_1] - a_{31}a_{32}p_2, \\
\check{l}_4 &= a_{32}\left[\left(a_{11} + a_{32} + a_{33} + p_2 + \frac{a_{42}\lambda_1}{p_2}\right)\left(a_{11} + a_{44} + p_2 + \frac{a_{42}\lambda_1}{p_2} - a_{31}\right)\right] \\
&\quad - j_{22(25)}\left(a_{11} + a_{33} + a_{44} + 2a_{32} + 2p_2 + \frac{2a_{42}\lambda_1}{p_2} - a_{22}\right).
\end{aligned}$$

Next, let $C_{23} = J_{23}C_{22}J_{23}^{-1}$, where the standard transformation matrix J_{23} is given by

$$J_{23} = \begin{pmatrix} a_{32}a_{35}p_2 & -\check{l}_1 & -\check{l}_2 & -\check{l}_3 & -\check{l}_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{23} = \begin{pmatrix} -(\rho+\check{l}_1) & -(\rho\check{l}_1+\check{l}_2) & -(\rho\check{l}_2+\check{l}_3) & -\rho\check{l}_3 & -\check{l}_4\left(p_2+a_{32}-a_{22}+\dfrac{a_{42}\lambda_1}{p_2}+\rho\right) \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & p_2+a_{32}-a_{22}+\dfrac{a_{42}\lambda_1}{p_2} \end{pmatrix}.$$

In this case, Eq. (4.4) becomes the following form

$$\begin{aligned} &(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)H^2(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)^T \\ &\quad + C_{23}(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)^T \\ &\quad + (J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)^T C_{23}^T = 0, \end{aligned}$$

that is,

$$H_0^2 + C_{23}\Delta_0 + \Delta_0 C_{23}^T = 0, \tag{4.11}$$

where $\Delta_0 = \varrho_7^{-2}(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)^T$, $\varrho_7 = a_{32}a_{35}p_2\sigma$. By solving Eq. (4.11), one gets

$$\Delta_0 = \begin{pmatrix} \dfrac{\rho^2(\check{l}_1\check{l}_2-\check{l}_3)+\check{l}_2(\rho\check{l}_2+\check{l}_3)}{\Sigma_{04}^*} & 0 & -\dfrac{\rho\check{l}_2+\check{l}_3}{\Sigma_{04}^*} & 0 & 0 \\ 0 & \dfrac{\rho\check{l}_2+\check{l}_3}{\Sigma_{04}^*} & 0 & -\dfrac{\rho+\check{l}_1}{\Sigma_{04}^*} & 0 \\ -\dfrac{\rho\check{l}_2+\check{l}_3}{\Sigma_{04}^*} & 0 & \dfrac{\rho+\check{l}_1}{\Sigma_{04}^*} & 0 & 0 \\ 0 & -\dfrac{\rho+\check{l}_1}{\Sigma_{04}^*} & 0 & \dfrac{\rho\check{l}_1(\rho+\check{l}_1)+\check{l}_1\check{l}_2-\check{l}_3}{\rho\check{l}_3\Sigma_{04}^*} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where

$$\Sigma_{04}^* = 2(\rho^3+\rho^2\check{l}_1+\rho\check{l}_2+\check{l}_3)(\check{l}_1\check{l}_2-\check{l}_3).$$

We can get that the matrix Δ_0 is positive semi-definite and its submatrix

$$\Delta_0^{(4)} = \begin{pmatrix} \frac{\rho^2(\check{l}_1\check{l}_2 - \check{l}_3) + \check{l}_2(\rho\check{l}_2 + \check{l}_3)}{\Sigma_{04}^*} & 0 & -\frac{\rho\check{l}_2 + \check{l}_3}{\Sigma_{04}^*} & 0 \\ 0 & \frac{\rho\check{l}_2 + \check{l}_3}{\Sigma_{04}^*} & 0 & -\frac{\rho + \check{l}_1}{\Sigma_{04}^*} \\ -\frac{\rho\check{l}_2 + \check{l}_3}{\Sigma_{04}^*} & 0 & \frac{\rho + \check{l}_1}{\Sigma_{04}^*} & 0 \\ 0 & -\frac{\rho + \check{l}_1}{\Sigma_{04}^*} & 0 & \frac{\rho\check{l}_1(\rho + \check{l}_1) + \check{l}_1\check{l}_2 - \check{l}_3}{\rho\check{l}_3\Sigma_{04}^*} \end{pmatrix}$$

is positive definite. As a result, the matrix $\Sigma = \varrho_7^2(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)^{-1}$ $\Delta_0[(J_{23}J_{22}J_{19}J_{18}J_{17}J_3J_2J_1)^{-1}]^T$ is also positive semi-definite.

Case 2.2. If $\lambda_1 = 0$, let $C_{24} = J_{24}C_{18}J_{24}^{-1}$, where the standard transformation matrix J_{24} takes the form

$$J_{24} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -a_{32}p_2 & \frac{a_{32}p_2}{a_{42}}(a_{31} - a_{11} - a_{44} - p_2) & (a_{31} - a_{11})^2 & a_{32}(a_{11} + a_{22} - a_{31} - a_{32}) \\ 0 & 0 & \frac{a_{32}p_2}{a_{42}} & a_{31} - a_{11} & -a_{32} \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{24} = \begin{pmatrix} -\rho & 0 & 0 & 0 & 0 \\ a_{32}a_{35}p_2 & -\hat{l}_1 & -\hat{l}_2 & -\hat{l}_3 & -\hat{l}_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & p_2 + a_{32} - a_{22} \end{pmatrix},$$

where

$$\begin{aligned} \hat{l}_1 &= a_{11} + a_{33} + a_{44} + a_{32} + p_2, \\ \hat{l}_2 &= (a_{11} + a_{44} + p_2 - a_{31})(a_{11} + a_{32} + a_{33}) - a_{32}p_2 - (a_{31} - a_{11})^2, \\ \hat{l}_3 &= (a_{11} - a_{31})(a_{44} + p_2)(a_{31} + a_{32} + a_{33}) - a_{11}a_{32}p_2, \\ \hat{l}_4 &= a_{32}[(a_{22} - a_{32} - p_2)(a_{22} - a_{31} - 2a_{32} - a_{33} - a_{44} - p_2) + a_{44}(a_{31} + a_{32} + a_{33})]. \end{aligned}$$

Define $C_{25} = J_{25}C_{24}J_{25}^{-1}$, where the standardized transformation matrix J_{25} is given by

$$J_{25} = \begin{pmatrix} a_{32}a_{35}p_2 & -\hat{l}_1 & -\hat{l}_2 & -\hat{l}_3 & -\hat{l}_4 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then

$$C_{25} = \begin{pmatrix} -(\rho+\hat{l}_1) & -(\rho\hat{l}_1+\hat{l}_2) & -(\rho\hat{l}_2+\hat{l}_3) & -\rho\hat{l}_3 & -\hat{l}_4(p_2+a_{32}-a_{22}+\rho) \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & p_2+a_{32}-a_{22} \end{pmatrix}.$$

In this case, Eq. (4.4) is rewritten in the following form

$$\begin{aligned} &(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)H^2(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)^T \\ &\quad + C_{25}(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)^T \\ &\quad + (J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)^T C_{25}^T = 0, \end{aligned}$$

i.e.,

$$H_0^2 + C_{25}\tilde{\Delta}_0 + \tilde{\Delta}_0 C_{25}^T = 0, \tag{4.12}$$

where $\tilde{\Delta}_0 = \varrho_7^{-2}(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)\Sigma(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)^T$, $\varrho_7 = a_{32}a_{35}p_2\sigma$. By solving Eq. (4.12), we have

$$\tilde{\Delta}_0 = \begin{pmatrix} \frac{\rho^2(\hat{l}_1\hat{l}_2-\hat{l}_3)+\hat{l}_2(\rho\hat{l}_2+\hat{l}_3)}{\Sigma_{05}^*} & 0 & -\frac{\rho\hat{l}_2+\hat{l}_3}{\Sigma_{05}^*} & 0 & 0 \\ 0 & \frac{\rho\hat{l}_2+\hat{l}_3}{\Sigma_{05}^*} & 0 & -\frac{\rho+\hat{l}_1}{\Sigma_{05}^*} & 0 \\ -\frac{\rho\hat{l}_2+\hat{l}_3}{\Sigma_{05}^*} & 0 & \frac{\rho+\hat{l}_1}{\Sigma_{05}^*} & 0 & 0 \\ 0 & -\frac{\rho+\hat{l}_1}{\Sigma_{05}^*} & 0 & \frac{\rho\hat{l}_1(\rho+\hat{l}_1)+\hat{l}_1\hat{l}_2-\hat{l}_3}{\rho\hat{l}_3\Sigma_{05}^*} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

where

$$\Sigma_{05}^* = 2(\rho^3+\rho^2\hat{l}_1+\rho\hat{l}_2+\hat{l}_3)(\hat{l}_1\hat{l}_2-\hat{l}_3).$$

It is easy to obtain that the matrix $\tilde{\Delta}_0$ is positive semi-definite and its submatrix

$$\tilde{\Delta}_0^{(4)} = \begin{pmatrix} \frac{\rho^2(\hat{l}_1\hat{l}_2 - \hat{l}_3) + \hat{l}_2(\rho\hat{l}_2 + \hat{l}_3)}{\Sigma_{05}^*} & 0 & -\frac{\rho\hat{l}_2 + \hat{l}_3}{\Sigma_{05}^*} & 0 \\ 0 & \frac{\rho\hat{l}_2 + \hat{l}_3}{\Sigma_{05}^*} & 0 & -\frac{\rho + \hat{l}_1}{\Sigma_{05}^*} \\ -\frac{\rho\hat{l}_2 + \hat{l}_3}{\Sigma_{05}^*} & 0 & \frac{\rho + \hat{l}_1}{\Sigma_{05}^*} & 0 \\ 0 & -\frac{\rho + \hat{l}_1}{\Sigma_{05}^*} & 0 & \frac{\rho\hat{l}_1(\rho + \hat{l}_1) + \hat{l}_1\hat{l}_2 - \hat{l}_3}{\rho\hat{l}_3\Sigma_{05}^*} \end{pmatrix}$$

is positive definite. Thereby, the matrix $\Sigma = \varrho_7^2(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)^{-1}\tilde{\Delta}_0[(J_{25}J_{24}J_{18}J_{17}J_3J_2J_1)^{-1}]^T$ is also positive semi-definite. The proof is confirmed. □

5 Extinction

In this section, we will present sufficient criteria for extinction exponentially of infected monocytes and free virus particles. Define

$$\mathcal{R}_0^E = \mathcal{R}_0 + \frac{[\mathcal{R}_0(\gamma + \bar{b}S^0) - \bar{b}S^0](e^{\frac{\sigma^2}{\rho}} - 2e^{\frac{\sigma^2}{4\rho}} + 1)^{\frac{1}{2}}}{\min\{\beta + \nu Z^0, \gamma + \bar{b}S^0\}},$$

where

$$\mathcal{R}_0 = \sqrt{\frac{mb\mu\delta}{(\alpha\gamma + b\mu)(\beta\delta + \nu\eta)}},\ S^0 = \frac{\mu}{\alpha},\ Z^0 = \frac{\eta}{\delta}.$$

Theorem 5.1 *Let $(S(t), I(t), V(t), Z(t), x(t))^T$ be the solution of system (1.4) with any initial value $(S(0), I(0), V(0), Z(0), x(0))^T \in \Xi$. If $\bar{b}S^0/(\gamma + \bar{b}S^0) < \mathcal{R}_0 < 1$ and $\mathcal{R}_0^E < 1$, then for almost $\omega \in \Omega$, we have*

$$\begin{aligned} &\limsup_{t\to\infty} \frac{1}{t} \ln\left(\frac{\mathcal{R}_0}{\beta + \nu Z^0} I(t) + \frac{\bar{b}S^0}{(\beta + \nu Z^0)(\gamma + \bar{b}S^0)} V(t)\right) \\ &\quad \le -\min\{\beta + \nu Z^0, \gamma + \bar{b}S^0\}(1 - \mathcal{R}_0^E) < 0 \ a.s., \end{aligned}$$

which shows the infected monocytes and free virus particles will go to extinction exponentially in a long term, i.e.,

$$\lim_{t\to\infty} I(t) = 0 \ and \ \lim_{t\to\infty} V(t) = 0 \ a.s.$$

Proof In light of Theorem 1.4 of Berman and Plemmons (1979), p. 27, we can get that there is a left positive eigenvector of a two-dimensional matrix

$$D = \begin{pmatrix} 0 & \dfrac{\bar{b}\mu\delta}{\alpha(\beta\delta+\nu\eta)} \\ \dfrac{m\alpha}{\alpha\gamma+\bar{b}\mu} & 0 \end{pmatrix}$$

corresponding to $\mathcal{R}_0$, which is given by

$$\mathcal{R}_0(\omega_1,\omega_2) = (\omega_1,\omega_2)D.$$

Define a C^2-function $\mathbb{W} : \mathbb{R}_+^2 \to \mathbb{R}_+$ as follows

$$\mathbb{W}(I,V) = \xi_1 I + \xi_2 V,$$

where

$$\xi_1 = \frac{\omega_1}{\beta+\nu Z^0},\ \xi_2 = \frac{\omega_2}{\gamma+\bar{b}S^0}.$$

Then applying Itô's formula (Mao 1997) to $\ln\mathbb{W}$ leads to that

$$\begin{aligned}
\frac{\mathrm{d}(\ln\mathbb{W})}{\mathrm{d}t} &= \frac{1}{\mathbb{W}}[\xi_1(bSV-\beta I-\nu IZ)+\xi_2(mI-\gamma V-bSV)] \\
&\le \frac{1}{\mathbb{W}}[\xi_1(\bar{b}S^0V-\beta I-\nu Z^0I)+\xi_2(mI-\gamma V-\bar{b}S^0V)] \\
&\quad + \frac{(\xi_1-\xi_2)S^0(b-\bar{b})V}{\mathbb{W}} + \frac{b(\xi_1-\xi_2)(S-S^0)V}{\mathbb{W}} \\
&\le \frac{1}{\mathbb{W}}\left[\frac{\omega_1}{\beta+\nu Z^0}(\bar{b}S^0V-\beta I-\nu Z^0I)+\frac{\omega_2}{\gamma+\bar{b}S^0}(mI-\gamma V-\bar{b}S^0V)\right] \\
&\quad + \frac{S^0(\xi_1-\xi_2)}{\xi_2}|b-\bar{b}| \\
&= \frac{1}{\mathbb{W}}(\omega_1,\omega_2)[D(I,V)^T-(I,V)^T]+\frac{S^0(\xi_1-\xi_2)}{\xi_2}|b-\bar{b}| \\
&= \frac{1}{\mathbb{W}}(\mathcal{R}_0-1)[(\beta+\nu Z^0)\xi_1 I+(\gamma+\bar{b}S^0)\xi_2 V]+\frac{\mathcal{R}_0(\gamma+\bar{b}S^0)-\bar{b}S^0}{\bar{b}}|b-\bar{b}| \\
&\le -\min\{\beta+\nu Z^0,\gamma+\bar{b}S^0\}(1-\mathcal{R}_0)+\frac{\mathcal{R}_0(\gamma+\bar{b}S^0)-\bar{b}S^0}{\bar{b}}|b-\bar{b}|.
\end{aligned} \tag{5.1}$$

Integrating both sides of (5.1) from 0 to t and then dividing by t yields that

$$\begin{aligned}
\frac{\ln\mathbb{W}(t)}{t} &\le \frac{\ln\mathbb{W}(0)}{t} - \min\{\beta+\nu Z^0,\gamma+\bar{b}S^0\}(1-\mathcal{R}_0) \\
&\quad + \frac{\mathcal{R}_0(\gamma+\bar{b}S^0)-\bar{b}S^0}{\bar{b}t}\int_0^t |b(s)-\bar{b}|\mathrm{d}s.
\end{aligned} \tag{5.2}$$

On the basis of the ergodicity of x and the strong law of large numbers (Lipster 1980), one has

$$\begin{aligned}\lim_{t\to\infty}\frac{1}{t}\int_0^t |e^{x(s)}-e^{\bar{x}}|\mathrm{d}s &= \int_{-\infty}^{\infty}|e^y-e^{\bar{x}}|\frac{\sqrt{\rho}}{\sqrt{\pi}\sigma}e^{-\frac{\rho(y-\bar{x})^2}{\sigma^2}}\mathrm{d}y\\ &\le \left(\int_{-\infty}^{\infty}(e^y-e^{\bar{x}})^2\frac{\sqrt{\rho}}{\sqrt{\pi}\sigma}e^{-\frac{\rho(y-\bar{x})^2}{\sigma^2}}\mathrm{d}y\right)^{\frac{1}{2}}\\ &\quad\left(\int_{-\infty}^{\infty}1^2\frac{\sqrt{\rho}}{\sqrt{\pi}\sigma}e^{-\frac{\rho(y-\bar{x})^2}{\sigma^2}}\mathrm{d}y\right)^{\frac{1}{2}}\\ &= \left(\int_{-\infty}^{\infty}(e^y-e^{\bar{x}})^2\frac{\sqrt{\rho}}{\sqrt{\pi}\sigma}e^{-\frac{\rho(y-\bar{x})^2}{\sigma^2}}\mathrm{d}y\right)^{\frac{1}{2}}\\ &= (e^{\frac{\sigma^2}{\rho}+2\bar{x}}-2e^{\frac{\sigma^2}{4\rho}+2\bar{x}}+e^{2\bar{x}})^{\frac{1}{2}}\\ &= \bar{b}(e^{\frac{\sigma^2}{\rho}}-2e^{\frac{\sigma^2}{4\rho}}+1)^{\frac{1}{2}}.\end{aligned} \tag{5.3}$$

Taking the superior limit on both sides of (5.2) and combining with (5.3) leads to that

$$\begin{aligned}\limsup_{t\to\infty}\frac{\ln\mathbb{W}(t)}{t} &\le -\min\{\beta+\nu Z^0,\gamma+\bar{b}S^0\}(1-\mathcal{R}_0)+[\mathcal{R}_0(\gamma+\bar{b}S^0)-\bar{b}S^0]\\ &\quad(e^{\frac{\sigma^2}{\rho}}-2e^{\frac{\sigma^2}{4\rho}}+1)^{\frac{1}{2}}\\ &:= -\min\{\beta+\nu Z^0,\gamma+\bar{b}S^0\}(1-\mathcal{R}_0^E)\text{ a.s.},\end{aligned}$$

where

$$\mathcal{R}_0^E=\mathcal{R}_0+\frac{[\mathcal{R}_0(\gamma+\bar{b}S^0)-\bar{b}S^0](e^{\frac{\sigma^2}{\rho}}-2e^{\frac{\sigma^2}{4\rho}}+1)^{\frac{1}{2}}}{\min\{\beta+\nu Z^0,\gamma+\bar{b}S^0\}}.$$

Accordingly, if $\mathcal{R}_0^E<1$, one can conclude that

$$\lim_{t\to\infty}I(t)=0\text{ and }\lim_{t\to\infty}V(t)=0\text{ a.s.},$$

which indicates that the infected monocytes and free virus particles will go to extinction exponentially a.s. The proof is confirmed. □

6 Numerical Simulations

In this section, we will provide several examples together with some numerical simulations to confirm our theoretical findings. We numerically simulate the solution of system (1.4) and take the initial value $(S(0), I(0), V(0), Z(0), x(0))^T = (1000, 1000, 1100, 1100, 2)^T$. By adopting the Milstein's higher-order method

Table 2 List of parameter values of system (1.4)

Parameter	Value	Source
μ	80	Nuraini et al. (2009)
$\bar{b}$	0.001	Nuraini et al. (2009)
θ	0.0005	Clapham et al. (2014)
α	0.2	Patel et al. (2017)
β	0.25	Patel et al. (2017)
m	20	Nuraini et al. (2009)
γ	0.8	Nuraini et al. (2009)
η	0.265	Nuraini et al. (2009)
k	0.0624	Ben-Shachar and Koelle (2015)
δ	1/180	Nuraini et al. (2009)
ρ	0.1	Estimated

involved in Higham (2001), one can get the discretization equations of system (1.4) as follows:

$$\begin{cases} S^{j+1} = S^j + (\mu - \alpha S^j - e^{x^j} S^j V^j)\Delta t, \\ I^{j+1} = I^j + (e^{x^j} S^j V^j - \beta I^j - \nu I^j Z^j)\Delta t, \\ V^{j+1} = V^j + (m I^j - \gamma V^j - e^{x^j} S^j V^j)\Delta t, \\ Z^{j+1} = Z^j + (\eta + k I^j + \theta I^j Z^j - \delta Z^j)\Delta t, \\ x^{j+1} = x^j + \rho(\bar{x} - x^j)\Delta t + \sigma\sqrt{\Delta t}\varepsilon_j, \end{cases} \tag{6.1}$$

where $(S^j, I^j, V^j, Z^j, x^j)^T$ is the corresponding value of the jth iteration of (6.1). Δt is the time increment and it is positive, ε_j are mutually independent random variables which follow the Gaussian distribution $\mathbb{N}(0, 1)$ for $j = 1, \ldots, n$ and n is the total number of iterations. We choose real parameter values from the previous literature, and all the values of biological parameters are shown in Table 2.

Example 6.1 (*Stationary distribution*) To illustrate the existence of a stationary distribution of system (1.4), we select $\nu = 0.001$ (Clapham et al. 2014), $\sigma^2 = 0.001$ and the other parameter values are given in Table 2. Then we calculate that

$$\mathcal{R}_0^S = \frac{m\bar{b}\mu\delta e^{\frac{\sigma^2}{12\rho}}}{(\alpha\gamma + \bar{b}\mu e^{\frac{\sigma^2}{4\rho}})(\beta\delta + \nu\eta)} \approx 22.3939 > 1.$$

In light of Theorem 3.1, the solution $(S(t), I(t), V(t), Z(t), x(t))^T$ of system (1.4) admits a stationary distribution $\pi(\cdot)$ which indicates the disease will be prevalent with probability one. Figure 1 validates this.

Example 6.2 (*Probability density function*) To study the existence of probability density around the quasi-endemic equilibrium E^*, we choose $\nu = 0.001$ (Clapham et al.

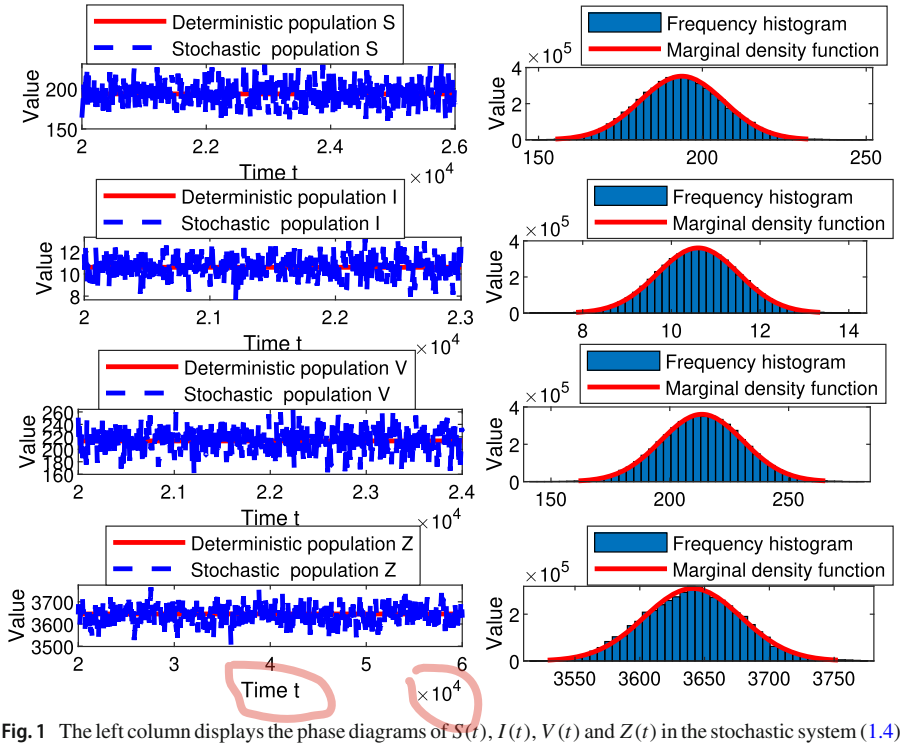

Fig. 1 The left column displays the phase diagrams of $S(t)$, $I(t)$, $V(t)$ and $Z(t)$ in the stochastic system (1.4) and their corresponding deterministic system (1.1). The right pictures show the corresponding frequency histograms and marginal density functions

2014) and the other parameters are shown in Table 2. One can compute that $E^* = (S^*, I^*, V^*, Z^*, \bar{x})^T = (193.4973, 10.6027, 213.4424, 3645.2764, -6.9078)^T$ and

$$\mathcal{R}_0^P = \sqrt{\frac{m\bar{b}\mu\delta}{(\alpha\gamma + \bar{b}\mu)(\beta\delta + \nu\eta)}} \approx 4.7322 > 1.$$

As a result, on the basis of Theorem 4.1, one can get that the solution $(S(t), I(t), V(t), Z(t), x(t))^T$ of system (1.4) has a Gaussian probability density $\Phi(S, I, V, Z, x) \sim \mathbb{N}(E^*, \Sigma)$, where

$$\Sigma = \begin{pmatrix} 167.2641 & -8.5797 & -194.3918 & -151.8745 & -0.7637 \\ -8.5797 & 0.8451 & 14.8288 & 0.1919 & 0.0546 \\ -194.3918 & 14.8288 & 300.4066 & 21.9302 & 0.9592 \\ -151.8745 & 0.1919 & 21.9302 & 1422.8373 & 1.0269 \\ -0.7637 & 0.0546 & 0.9592 & 1.0269 & 0.005 \end{pmatrix}.$$

Therefore, one can get the following marginal density functions:

$$\Phi_1(S) = 0.0308e^{-0.0030(S-193.4973)^2}, \ \Phi_2(I) = 0.4340e^{-0.5917(I-10.6027)^2},$$

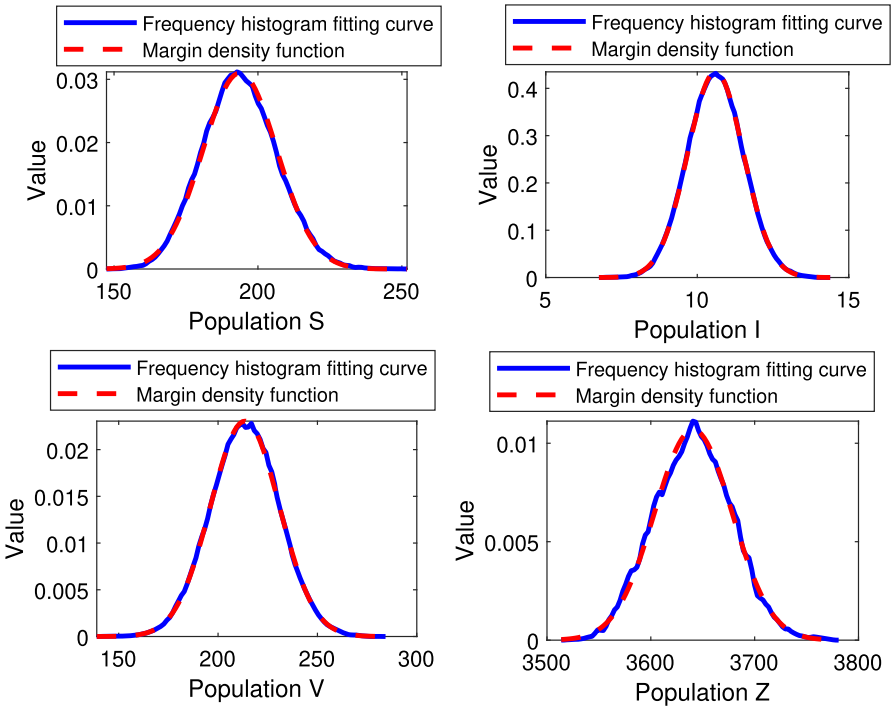

Fig. 2 The column displays the fitted plots of the corresponding frequency histograms and marginal density functions, respectively

$$\Phi_3(V) = 0.0230e^{-0.0017(V-213.4424)^2}, \ \Phi_4(Z) = 0.0106e^{-0.0004(Z-3645.2764)^2}.$$

Figure 2 shows the corresponding marginal density functions and frequency histograms with 10,000,000 iteration points, respectively.

Example 6.3 (*Extinction*) To verify the extinction of infected monocytes and free viruses, we choose $\nu = 0.15$, $\sigma^2 = 0.0001$ and the other parameter values are given in Table 2. Then

$$\frac{\bar{b}S^0}{\gamma + \bar{b}S^0} \approx 0.3333 < 0.9488 \approx \mathcal{R}_0 \approx 0.9488 < 1$$

and

$$\mathcal{R}_0^E = \mathcal{R}_0 + \frac{[\mathcal{R}_0(\gamma + \bar{b}S^0) - \bar{b}S^0](e^{\frac{\sigma^2}{\rho}} - 2e^{\frac{\sigma^2}{4\rho}} + 1)^{\frac{1}{2}}}{\min\{\beta + \nu Z^0, \gamma + \bar{b}S^0\}} \approx 0.9626 < 1.$$

Accordingly, by Theorem 5.1, the infected monocytes and free viruses will be extinct in a long term. Figure 3 shows this.

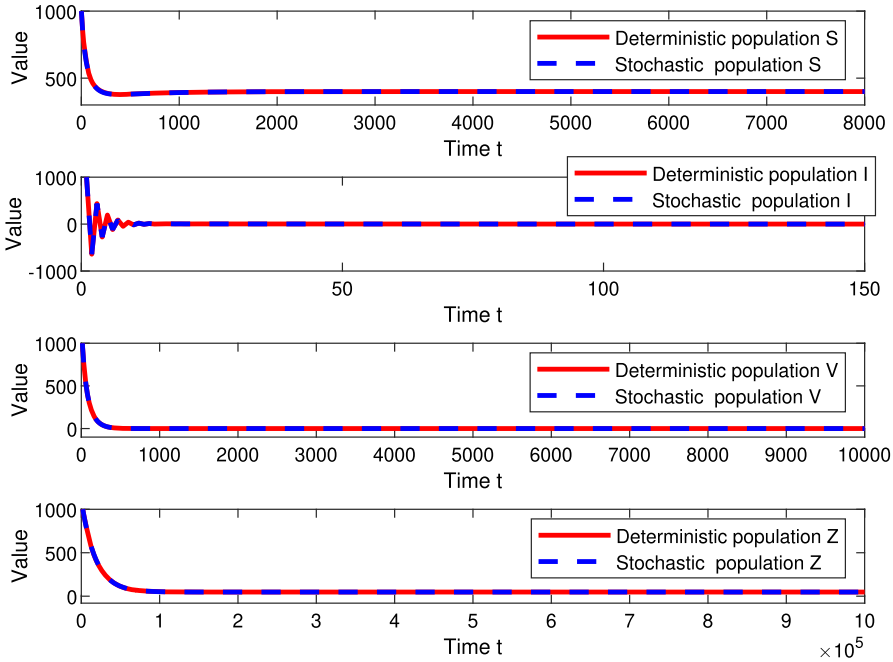

Fig. 3 Computer simulations for the susceptible monocyte population S; the infected monocyte population I; the viral load V and the immune cells Z of (1.4) with the noise intensity $\sigma^2 = 0.0001$

7 Conclusion

It is worth noting that environmental noise plays an important role in the transmission of infectious diseases. In this context, we consider the potential impact of environmental variability in the contact rate by employing the Ornstein–Uhlenbeck process in natural logarithmic form due to the biological significance. This technique serves as an improved means of introducing environmental noise into population dynamic models in mathematically reasonable hypotheses, as previously illustrated in Allen (2016). On the basis of this fact, we give and consider a stochastic dengue model with Ornstein–Uhlenbeck process to obtain new insights into the transmission dynamics of the dengue fever in a population. The principal mathematical findings of our paper are listed as follows:

- If

$$\mathcal{R}_0^S = \frac{m\bar{b}\mu\delta e^{\frac{\sigma^2}{12\rho}}}{(\alpha\gamma + \bar{b}\mu e^{\frac{\sigma^2}{4\rho}})(\beta\delta + \nu\eta)} > 1,$$

then the stochastic system (1.4) has at least one stationary distribution $\pi(\cdot)$ on Ξ.

- If

$$\mathcal{R}_0^P = \sqrt{\frac{m\bar{b}\mu\delta}{(\alpha\gamma + \bar{b}\mu)(\beta\delta + \nu\eta)}} > 1,$$

then the stationary solution $(S(t), I(t), V(t), Z(t), x(t))^T$ of system (1.4) near E^* follows a normal distribution $\mathbb{N}(E^*, \Sigma)$.

- If

$$\frac{\bar{b}S^0}{\gamma + \bar{b}S^0} < \mathcal{R}_0 < 1 \text{ and } \mathcal{R}_0^E = \mathcal{R}_0 + \frac{[\mathcal{R}_0(\gamma + \bar{b}S^0) - \bar{b}S^0](e^{\frac{\sigma^2}{\rho}} - 2e^{\frac{\sigma^2}{4\rho}} + 1)^{\frac{1}{2}}}{\min\{\beta + \nu Z^0, \gamma + \bar{b}S^0\}} < 1,$$

then the infected monocytes and free virus particles will go to extinction exponentially in a long term, i.e.,

$$\lim_{t\to\infty} I(t) = 0 \text{ and } \lim_{t\to\infty} V(t) = 0 \text{ a.s.}$$

In addition, on the one hand, according to Theorems 3.1 and 4.1, we can get that the solution of system (1.4) is asymptotically stable in distribution, that is, the dengue fever will be persistent in human beings with probability one. And if $\sigma = 0$, then $\mathcal{R}_0^S = \mathcal{R}_0$. Therefore, $\mathcal{R}_0^S > 1$ is considered to be a unified criterion for persistence of dengue fever in the deterministic system (1.1) and the corresponding stochastic version (1.4). One the other hand, in view of the expressions of $\mathcal{R}_0^E$ and $\mathcal{R}_0$, it is easy to see that $\mathcal{R}_0 \leq \mathcal{R}_0^E$ and the equals sign holds if and only if $\sigma = 0$. Hence, $\mathcal{R}_0^E < 1$ can be seen as a unified criterion for extinction of dengue fever in systems (1.1) and (1.4). Furthermore, it is unable to determine the size of $\mathcal{R}_0^S$ and $\mathcal{R}_0$. This indicates that environmental noises have great destabilizing impacts on the transmission dynamics of dengue fever. Our analysis has highlighted the influences of certain parameters on the dynamics of system (1.4), which can be employed to develop effective strategies for disease control. On the basis of the results of Theorems 3.1 and 5.1, we can conclude that an increase in the natural death rate of viral load or the contact rate between susceptible monocytes and viral load corresponds to a lower risk of infection. Regarding the expressions of $\mathcal{R}_0^S$ and $\mathcal{R}_0^E$, we have found that environmental noises have great effects on the transmission dynamics of dengue fever. Meanwhile, we have found that decreasing the noise intensity is also an effective strategy for controlling the spread of this type of disease and reducing the number of infected monocytes and viral load. In other words, increasing the noise intensity is not conducive to the elimination of dengue fever.

Finally, according to the results of numerical simulation, one can find a strange phenomenon: as the noise intensity σ increases, disease that was previously extinct will become persistent, with all other parameters remaining constant. This is very different from the existing conclusions of the linear white noise perturbation. One interesting problem is that when we keep a decrease in the mean reversion rate ρ and

retain the other parameters constant, what changes will appear in the dynamics of infectious diseases? These topics are currently underway.

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Data Availability Data sharing is not applicable to this article as no datasets are generated or analyzed during the current study.

Declarations

Conflict of interest The author declares that there is no conflict of interest.

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